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Speed reducer

Abstract

A speed reducer relating to one aspect of the present disclosure includes: an internal gear having a casing and a plurality of outer pins, where the casing surrounds a main axis, and the outer pins are rotatably arranged in pin grooves provided on an inner periphery of the casing; an external gear meshing with the internal gear; an eccentric body for oscillating the external gear; a carrier rotatable relative to the casing; and a main bearing having an inner sliding surface and an outer sliding surface, where the inner sliding surface is rotatable integrally with the casing, and the outer sliding surface is rotatable integrally with the carrier. One of the inner and outer sliding surfaces is made of resin, and the other of the inner and outer sliding surfaces is made of a thermally conductive material that is more wear-resistant than the resin.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Divisional Application of U.S. Ser. No. 17/590,552, filed Feb. 1, 2022, which is based on and claims the

benefit of priority from Japanese Patent Application Serial No. 2021-016783, filed on Feb. 4, 2021. The contents of each of which is hereby incorporated by reference in entirety.

TECHNICAL FIELD

(1) The present disclosure relates to a speed reducer.

BACKGROUND

(2) In the conventional art, eccentric oscillation speed reducers are known that are provided between two members and configured to reduce the number of rotations with a predetermined reduction ratio. Such eccentric oscillation speed reducers include an outer cylinder fixedly attached to one of the members and a carrier provided within the outer cylinder and fixedly attached to the other of the members. The carrier is rotatable relative to the outer cylinder in response to oscillatory rotation of oscillating gears attached to eccentric portions of crankshafts.

(3) Recent years have seen a change in the environment in which robots are used. There has been a trend toward further reduction in size and weight of robots, which in turn requires reduction in size and weight of speed reducers. As disclosed in Japanese Patent Application Publication No. 2018-17362 (“the '362 Publication”), the parts of the speed reducers can be reduced in weight by, for example, using resin parts.

(4) The technology disclosed in the '362 Publication, however, only achieve limited success in weight reduction. Furthermore, when the technology disclosed in the '362 Publication is employed and many resin parts are used to achieve weight reduction, the resin parts may result in a lack of strength and accordingly a higher defect rate. In addition, the speed reducers may experience a temperature rise while they are in use. In this case, the resin parts may cause deformation, malfunction or other problems in the speed reducers.

SUMMARY

(5) The present disclosure aims to provide speed reducers that can achieve both weight reduction and rigidity enhancement. (1) One aspect of the present disclosure provides a speed reducer including an internal gear having a casing (corresponding to an internal gear body) and a plurality of outer pins, where the casing surrounds a main axis, and the outer pins are rotatably arranged in pin grooves provided on an inner periphery of the casing, an external gear meshing with the internal gear, an eccentric body for oscillating the external gear; a carrier rotatable relative to the casing, and a main bearing having an inner sliding surface and an outer sliding surface, where the inner sliding surface is rotatable integrally with the casing, and the outer sliding surface is rotatable integrally with the carrier. One of the inner and outer sliding surfaces is made of resin, and the other of the inner and outer sliding surfaces is made of a thermally conductive material that is more wear-resistant than the resin.

(6) According to the speed reducer relating to one aspect of the present disclosure, the outer pins are provided between the casing and the external gear. This can contribute to improve the rigidity and heat dissipation while the reduction in size and weight is still successful. As a result, the speed reducer can effectively avoid failures. (2) In the speed reducer described in (1), the main bearing may be configured such that at least the inner sliding surface is formed on an inner peripheral surface of the casing or the outer sliding surface is formed on an outer peripheral surface of the carrier. (3) In the speed reducer described in (2), the carrier may be made of the resin, and the outer sliding surface may be formed on the outer peripheral surface of the carrier. (4) In the speed reducer described in (3), the inner sliding surface may be made of the thermally conductive material, and, in a direction extending along the main axis, the inner sliding surface may be positioned to overlap the outer pins. (5) One aspect of the present disclosure provides a speed reducer including an internal gear having a casing (corresponding to an internal gear body) and a plurality of outer pins, where the casing surrounds a main axis, and the outer pins are rotatably arranged in pin grooves provided on an inner periphery of the casing, an external gear meshing with the internal gear, an eccentric body for oscillating the external gear, a carrier rotatable relative

to the casing, and a main bearing having an inner sliding surface and an outer sliding surface, where the inner sliding surface is formed on an inner periphery of a metal ring rotatable integrally with the casing, and the outer sliding surface is formed on an outer peripheral surface of the carrier. The casing and the carrier are made of resin. In a direction extending along the main axis, the metal ring and the outer pins are positioned to overlap each other.

(7) According to the speed reducer relating to one aspect of the present disclosure, the outer pins are provided between the casing and the external gear, and the outer pins and the metal rings extend in the direction extending along the main axis such that they wrap each other. This can contribute to further improve the rigidity while the reduction in size and weight is still successful. As a result, the speed reducer can effectively avoid failures. (6) One aspect of the present disclosure provides a speed reducer including an internal gear provided on an inner periphery of a casing surrounding a main axis, an external gear meshing with the internal gear, an eccentric body for oscillating the external gear, a carrier rotatable relative to the casing, and a main bearing having an inner sliding surface and an outer sliding surface, where the inner sliding surface is rotatable integrally with the casing, and the outer sliding surface is rotatable integrally with the carrier. One of the inner and outer sliding surfaces is made of resin, and the other of the inner and outer sliding surfaces is made of a thermally conductive material that is more wear-resistant than the resin. At least one of the inner sliding surface or the outer sliding surface is inclined such that a radial dimension thereof relative to the main axis increases or decreases in a direction extending along the main axis.

(8) According to the speed reducer relating to one aspect of the present disclosure, the plain bearings have a large contact area at the bearing surface, which can result in more stable operation.

(7) In the speed reducer described in (6), the inner and outer sliding surfaces may have a projection as a crush allowance. (8) In the speed reducer described in any of (1) to (7), the inner and outer sliding surfaces may have a groove formed therein that is not in communication with an internal space having the external gear housed therein. (9) One aspect of the present disclosure provides a speed reducer including an internal gear provided on an inner periphery of a casing surrounding a main axis, an external gear meshing with the internal gear, an eccentric body for oscillating the external gear, a carrier rotatable relative to the casing, and a main bearing having an inner sliding surface and an outer sliding surface, where the inner sliding surface is formed on an inner peripheral surface of the casing, and the outer sliding surface is formed on an outer peripheral surface of the carrier. The casing is made of a metal, and the carrier is made of resin. The inner and outer sliding surfaces are inclined such that a radial dimension thereof relative to the main axis increases in a direction extending along the main axis away from the external gear. The inner and outer sliding surfaces have a projection as a crush allowance. The inner and outer sliding surfaces have a groove formed therein that is not in communication with an internal space having the external gear housed therein.

(9) According to the speed reducer relating to one aspect of the present disclosure, the inner and outer sliding surfaces can catch dust and excessive grease in the groove. This can prevent the dust and the like from affecting the sliding on the inner and outer sliding surfaces. In addition, the dust and the like can be prevented from entering the speed reducer. (10) One aspect of the present disclosure provides an eccentric oscillation speed reducer for transmitting a driving force between a first member and a second member. The eccentric oscillation speed reducer is configured to change the number of rotations at a predetermined ratio. The eccentric oscillation speed reducer includes an eccentric portion, an external gear having external teeth and an insertion hole into which the eccentric portion is inserted, a casing attachable to one of the first and second members, a carrier attachable to the other of the first and second members, and a main bearing having an inner sliding surface and an outer sliding surface, where the inner sliding surface is rotatable integrally with the casing, and the outer sliding surface is rotatable integrally with the carrier. The casing has internal teeth meshing with the external teeth of the external gear. The carrier is placed within the casing in a radial direction while holding the external gear. The casing and the carrier are rotatable relative to

each other in a concentric manner in response to oscillation of the external gear resulting from rotation of the eccentric portion. One of the inner and outer sliding surfaces is made of resin, and the other of the inner and outer sliding surfaces is made of a thermally conductive material that is more wear-resistant than the resin.

(10) According to the speed reducer relating to one aspect of the present disclosure, the outer pins are provided between the casing and the external gear. This can contribute to improve the rigidity and heat dissipation while the reduction in size and weight is still successful. As a result, the speed reducer can effectively avoid failures. (11) One aspect of the present disclosure provides an eccentric oscillation speed reducer including two or more eccentric bodies and external gears respectively corresponding to the eccentric bodies. The eccentric oscillation speed reducer includes a casing having an internal gear on an inner periphery thereof, where the internal gear meshes with the external gears, a carrier rotatable relative to the casing and rotatable relative to the eccentric bodies, and a main bearing having an inner sliding surface and an outer sliding surface, where the inner sliding surface is formed on an inner peripheral surface of the casing, and the outer sliding surface is formed on an outer peripheral surface of the carrier. One of the inner and outer sliding surfaces is made of resin, and the other of the inner and outer sliding surfaces is made of a thermally conductive material that is more wear-resistant than the resin.

(11) According to the speed reducer relating to one aspect of the present disclosure, the outer pins are provided between the casing and the external gear. This can contribute to improve the rigidity and heat dissipation while the reduction in size and weight is still successful. As a result, the speed reducer can effectively avoid failures. (12) One aspect of the present disclosure provides a speed reducer including an internal gear having a casing (corresponding to an internal gear body) and a plurality of outer pins, the casing surrounding a main axis, the outer pins being rotatably arranged in pin grooves provided on an inner periphery of the casing, an external gear meshing with the internal gear, an eccentric body for oscillating the external gear, a carrier rotatable relative to the casing, and a main bearing having an inner surface and an outer surface, where the inner surface is rotatable integrally with the casing, the outer surface is rotatable integrally with the carrier, and the outer pins are sandwiched between the inner surface and the outer surface. The inner and outer surfaces are made of one of a resin and a thermally conductive material that is more wear-resistant than the resin, and the outer pins are made of the other.

(12) According to the speed reducer relating to one aspect of the present disclosure, the outer pins are provided between the casing and the external gear. This can contribute to improve the rigidity and heat dissipation while the reduction in size and weight is still successful. As a result, the speed reducer can effectively avoid failures. (13) In the speed reducer described in (12), the outer pins may have, at an end thereof (one or both of the ends), a diameter-enlarged portion having an enlarged diameter. (14) One aspect of the present disclosure provides a speed reducer including an internal gear having a casing (corresponding to an internal gear body) and a plurality of outer pins, where the casing surrounds a main axis, and the outer pins are rotatably arranged in pin grooves provided on an inner periphery of the casing, an external gear meshing with the internal gear, an eccentric body for oscillating the external gear, a carrier rotatable relative to the casing, a first main bearing having an inner surface and an outer surface, the inner surface being rotatable integrally with the casing, the outer surface being rotatable integrally with the carrier, the outer pins being sandwiched between the inner surface and the outer surface, and a second main bearing configured as a crossed roller bearing including a plurality of rollers disposed between V grooves having a V-shaped section facing each other. The inner and outer surfaces are made of one of resin and a thermally conductive material that is more wear-resistant than the resin, and the outer pins are made of the other. (15) One aspect of the present disclosure provides a speed reducer including an internal gear having a casing (corresponding to an internal gear body) and a plurality of outer pins, the casing surrounding a main axis, the outer pins being rotatably arranged in pin grooves provided on an inner periphery of the casing, an external gear meshing with the internal gear, an eccentric

body for oscillating the external gear, a carrier rotatable relative to the casing, and a main bearing having an inner surface and an outer surface, where the inner surface is rotatable integrally with the casing, the outer surface is rotatable integrally with the carrier, and the outer pins are sandwiched between the inner surface and the outer surface. The inner and outer surfaces are made of resin. The outer pins are made of a metal. The outer pins have, at an end thereof, a diameter-enlarged portion having an enlarged diameter.

(13) According to the speed reducer relating to one aspect of the present disclosure, the outer pins are provided between the casing and the external gear. This can contribute to improve the rigidity and heat dissipation while the reduction in size and weight is still successful. As a result, the speed reducer can effectively avoid failures and operate more reliably.

(14) The present disclosure can effectively achieve improved rigidity and heat dissipation while accomplishing smaller size and lighter weight and also prevent failures and realize more reliable operation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a sectional view showing a speed reducer relating to a first embodiment of the present disclosure along a main axis.
- (2) FIG. 2 is a sectional view along a line II-II in FIG. 1.
- (3) FIG. 3 is a sectional view showing a speed reducer relating to a second embodiment of the present disclosure along a main axis.
- (4) FIG. 4 is a sectional view showing a speed reducer relating to a third embodiment of the present disclosure along a main axis.
- (5) FIG. 5 is a sectional view along a line V-V in FIG. 4.
- (6) FIG. 6 is a sectional view showing a speed reducer relating to a fourth embodiment of the present disclosure along a main axis.
- (7) FIG. 7 is an enlarged sectional view showing a speed reducer relating to a fifth embodiment of the present disclosure, more specifically, a region near a main bearing.
- (8) FIG. 8 is a sectional view showing a casing constituting an inner sliding surface of a main bearing of a speed reducer relating to a sixth embodiment of the present disclosure.
- (9) FIG. 9 is an enlarged sectional view showing the casing to indicate an example relation between a groove and a crush allowance of the inner sliding surface in the sixth embodiment.
- (10) FIG. 10 is a sectional view showing the casing to indicate an example relation between the groove and the crush allowance of the inner sliding surface in the sixth embodiment.
- (11) FIG. 11 is a sectional view showing a speed reducer relating to a seventh embodiment of the present disclosure along a main axis.
- (12) FIG. 12 is a sectional view along a line XII-XII in FIG. 11.
- (13) FIG. 13 is a sectional view showing a speed reducer relating to an eighth embodiment of the present disclosure along a main axis.
- (14) FIG. 14 is a sectional view showing a speed reducer relating to a ninth embodiment of the present disclosure along a main axis.
- (15) FIG. 15 is a sectional view along a line XV-XV in FIG. 14.
- (16) FIG. 16 is a sectional view showing a speed reducer relating to a tenth embodiment of the present disclosure along a main axis.
- (17) FIG. 17 is a sectional view showing a speed reducer relating to an eleventh embodiment of the present disclosure along a main axis.
- (18) FIG. 18 is a sectional view showing a speed reducer relating to a twelfth embodiment of the present disclosure along a main axis.

(19) FIG. **19** is a sectional view showing a speed reducer relating to a thirteenth embodiment of the present disclosure along a main axis.

(20) FIG. **20** is a sectional view showing a speed reducer relating to a fourteenth embodiment of the present disclosure along a main axis.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

First Embodiment

(21) The following describes a speed reducer relating to a first embodiment of the present disclosure with reference to the accompanying drawings. FIG. **1** is a sectional view showing the speed reducer relating to the first embodiment along a main axis. FIG. **2** is a sectional view along a line II-II in FIG. **1**. The dimensions of the components in the drawings are shown enlarged or reduced as appropriate for ease of understanding. In the drawings, some parts that are not important for explaining the first embodiment are not shown. In FIGS. **1** and **2**, the reference numeral **100** indicates the speed reducer. FIG. **2** only shows one of the two external gears **114** for ease of understanding.

(22) The speed reducer **100** relating to the first embodiment is an eccentric oscillation speed reducer having an external gear and an internal gear meshing with each other. Oscillating the external gear causes one of the internal and external gears to rotate on its own axis, so that the generated rotational component is output to a driven device through an output member.

(23) As shown in FIGS. **1** and **2**, the eccentric oscillation speed reducer **100** has an input shaft **112**, an external gear **114**, an internal gear **116**, a first carrier **118**, a second carrier **120**, a casing **122**, a first main bearing **124**, a second main bearing **126**, an inner pin **140** and a carrier pin **138**. Hereinafter, the expression “axial direction” refers to the direction extending along the central axis (the main axis) **1La** of the internal gear **116**, and the expressions “circumferential direction” and “radial direction” respectively refer to the circumferential and radial directions of the circle centered on the central axis **1La**. In the following description, for convenience, the input side refers to one side in the axial direction (the right side in FIG. **1**) and the non-input side or output side refers to the other side (the left side in FIG. **1**). Furthermore, in the following description, the first and second carriers **118** and **120** may be simply referred to as the carriers **118** and **120**. Likewise, the first and second main bearings **124** and **126** may be simply referred to as the main bearings **124** and **126**.

(24) The input shaft **112** is rotatable around the center line when acted upon by rotational power input from a drive source. The speed reducer **100** relating to the present embodiment is a center crank type, where the rotational center line of the input shaft **112** is aligned with the central axis **1La** of the internal gear **116**. The drive source is, for example, a motor, a gear motor or an engine.

(25) The input shaft **112** is an eccentric body having a plurality of eccentric portions **112a** for oscillating the external gear **114**. Having such a configuration, the input shaft (eccentric body) **112** is sometimes referred to as a crankshaft. The axial center of the eccentric portions **112a** is eccentric to the rotational center line of the input shaft **112**. In the present embodiment, the input shaft **112** includes two eccentric portions **112a** adjacent to each other. The adjacent eccentric portions **112a** are 180° out of phase in terms of eccentricity.

(26) The input shaft **112** is supported on the input side by the second carrier **120** via an input shaft bearing **134** and on the non-input side by the first carrier (shaft flange) **118** via an input shaft bearing **134**. The input shaft **112** is supported such that it is rotatable relative to the first carrier **118** and the second carrier (hold flange) **120**. The input shaft bearings **134** are not limited to any particular configurations, but are, for example, ball bearings with spherical rollers in the present embodiment.

(27) The internal gear **116** meshes with the external gear **114**. According to the present embodiment, the internal gear **116** includes an internal gear body **116a** integrated into the casing **122**, and outer pins (internal tooth pins) **117** disposed in pin grooves **116b** formed in the internal gear body **116a** and arranged at intervals in the circumferential direction. The outer pins **117** are

cylindrical pin members rotatably supported on the internal gear body **116a**. The outer pins **117** form the internal teeth of the internal gear **116**. The number of outer pins **117** (the number of internal teeth) of the internal gear **116** is slightly (in this example, by one) greater than the number of external teeth of the external gear **11**.

(28) Basically, the outer pins **117** are all shaped in the same manner. The radial dimension of the outer pins **117** remains constant along their entire length along the central axis **1La**. All of the outer pins **117** run parallel to the central axis **1La**. The outer pins **117** are all positioned in the same manner in the radial direction and in the axial direction along the central axis **1La**, while being spaced away from each other in the circumferential direction.

(29) The internal gear body **116a** and the casing **122**, which are integrated to each other, are made of resin. The internal gear body **116a** and the casing **122** can be made of a variety of resin. In this example, the internal gear body **116a** and the casing **122** are made of polyacetal (POM). The present embodiment, however, is not limited to such, and the internal gear body **116a** and the casing **122** may be made of other resin than POM, for example, polyetheretherketone (PEEK) and PAEK (polyaryletherketones).

(30) The resin used to make the internal gear body **116a**, the casing **122** and other constituent parts of the present embodiment may be resin containing reinforcing fibers such as glass or carbon fibers, resin without reinforcing fibers, lamination of a base material such as paper or fabric impregnated with resin. The resin used for the constituent parts of the present embodiment may be resin having thermally conductive fillers mixed therein.

(31) In the speed reducer **100**, the outer pins **117** may be made of a material that has a higher thermal conductivity [W/(m.Math.K)] and that is more resistant against wear than the resin material used for the internal gear body **116a**. In the following description, metals are introduced as an example thermally conductive material that is more resistant against wear than resin. As used herein, the term “metals” include thermally conductive materials that are more resistant against wear than the above-listed resin.

(32) The outer pins **117** can be made of any material as long as it exhibits higher resistance against wear and higher thermal conductivity than the resin used to make the internal gear body **116a** and can be made of metals, highly thermally conductive resin, non-metal materials and the like. The outer pins **117** may be, for example, made of resin having carbon nanotubes (CNTs), boron nitride nanotubes (BNNTs) mixed therein. The outer pins **117** relating to the present embodiment, shown in FIGS. **1** and **2**, may be made of iron-based metals such as bearing steel.

(33) The outer pins **117** can be solid or hollow. The outer pins **117** may have a multi-layered structure with a core material being wrapped in a surface material. For example, one of the core and surface materials of the multi-layered structure of the outer pins **117** may be made of iron-based metals, and the other may be made of a copper- or aluminum-based metal. Such a structure can eliminate the trade-off between the mechanical and thermal characteristics. As yet another example of the core and surface materials of the multi-layered structure of the outer pins **117**, one of them may be made of metals and the other may be made of resin. In a further different example, the outer pins **117** may be made of sintered metals.

(34) The speed reducer **100** has a plurality of external gears **114** provided in a one-to-one correspondence with the eccentric portions **112a**. The external gears **114** are rotatably supported on the eccentric portions **112a** via eccentric bearings **130**. As shown in FIG. **2**, the external gears **114** each have ten through holes that are arranged at equal intervals and offset from the central axis of the external gears **114**. Three of the ten through holes, which are at equal intervals of 120 degrees around the axis of the external gears **114**, serve as carrier pin holes **139**. The carrier pin holes **139** receive carrier pins **138**. The remaining seven through holes serve as inner pin holes **141**. The inner pin holes **141** receive inner pins **140**. The carrier pin holes **139** may have the same diameter as the inner pin holes **141**.

(35) The external gears **114** are made of resin, like the internal gear body **116a**. The external gears

114 can be made of a variety of resin. The external gears **114** are located closer to the input shaft **112** than is the internal gear body **116a**. The external gears **114** are made of, for example, PEEK. The present embodiment, however, is not limited to such, and the external gears **114** may be made of, for example, POM or other resins than PEEK.

(36) The carrier pin holes **139** and the inner pin holes **141** are at the same distance from the center in the radial direction and shaped like a circle having the same diameter. Corrugated teeth are formed on the outer periphery of the external gears **114**. The external gears **114** oscillate within a plane to which the central axis of the external gears **114** is normal, such that the corrugated teeth move while keeping touching the internal gear **116**. A gap is provided between the inner pins **140** and the inner pin holes **141** to absorb the oscillating component of the external gears **114**. The inner pins **140** partially touch the inner wall surface of the inner pin holes **141**.

(37) The carriers **118** and **120** sandwich the external gears **114** therebetween in the axial direction. The first carrier (shaft flange) **118** is located on the lateral portion of the external gears **114** on the non-input side. The second carrier (hold flange) **120** is located on the lateral portion of the external gears **114** on the input side. The first carrier (shaft flange) **118** is rotatably supported by the casing **122** via a first main bearing **124**. The second carrier (hold flange) **120** is rotatably supported by the casing **122** via a second main bearing **126**.

(38) The carriers **118** and **120** are shaped like a disc as a whole. The first carrier **118** rotatably supports the input shaft **112** via the input shaft bearings **134**. The second carrier **120** rotatably supports the input shaft **112** via the input shaft bearings **134**. The carrier pins **138** are coupled to the first carrier (shaft flange) **118** and to the second carrier (hold flange) **120** with bolts **138a**. The inner pins **140** are coupled to the first carrier (shaft flange) **118** and to the second carrier (hold flange) **120** via, for example, bolts **140a** made of iron-based metals.

(39) The first and second carriers **118** and **120** are coupled to each other via the carrier and inner pins **138** and **140**. The carrier and inner pins **138** and **140** are positioned off the axial center of the external gears **114** in the radial direction and penetrate the external gears **114** in the axial direction. In this example, the carrier and inner pins **138** and **140** are provided as separate parts from the carriers **118** and **120**. The present embodiment, however, is not limited to such. As another example, the carrier and inner pins **138** and **140** may be partly integrated into the carriers **118** and **120**.

(40) One of the first carrier **118** and the casing **122** serves as an output member for outputting rotational power to a driven device, and the other serves as a stationary member fixedly attached to an external member for supporting the speed reducer **100**. The output member is rotatably supported on the stationary member via the main bearings **124** and **126**. The non-input-side end surface of the first carrier **118** may have a driven member, which is to be rotated by the speed reducer **100**, coupled thereto with bolts or the like. Alternatively, the outer peripheral flange of the casing **122** may have a driven member, which is to be rotated by the speed reducer **100**, coupled thereto with bolts or the like.

(41) The casing **122** is shaped like a hollow tube as a whole. The internal gear **116** is provided on the inner periphery of the casing **122**. A flange or the like may be provided on the outer periphery of the casing **122**. The casing **122** has a first metal ring **144** located on the non-input side of the casing **122** and a second metal ring **145** located on the input side of the casing **122**. The first and second metal rings **144** and **145** are fixedly attached to and integrated into the casing **122**. The first and second metal rings **144** and **145** may be fixedly attached to the casing **122** with a plurality of bolts arranged next to each other in the circumferential direction.

(42) In the following description, the first and second metal rings **144** and **145** may be simply referred to as the metal rings **144** and **145**.

(43) The casing **122** has a depression receiving therein the first metal ring **144**. The first metal ring **144** protrudes in the axial direction beyond the depression. The first metal ring **144** serves as the outer ring of the first main bearing **124**. The first metal ring **144** may have the outer periphery flush

with the outer periphery of the casing **122**.

(44) The first metal ring **144**, which serves as the outer ring of the first main bearing **124**, is located on the output side of the casing **122**. The first metal ring **144** is exposed on the output side of the casing **122**. The portion of the first metal ring **144** that protrudes in the axial direction beyond the depression of the casing **122** constitutes an inner sliding surface **148** serving as the outer ring. The inner sliding surface **148** is in contact with the first carrier (shaft flange) **118**. The first metal ring **144** is firmly fitted onto the casing **122**, and the fit can be, for example, a clearance fit, an interference fit or a transition fit. The clearance may be determined by the difference in thermal expansion coefficient.

(45) The casing **122** has a depression for receiving therein the second metal ring **145**. The second metal ring **145** protrudes in the axial direction beyond the depression. The second metal ring **145** serves as the outer ring of the second main bearing **126**. The second metal ring **145** may have the outer periphery flush with the outer periphery of the casing **122**.

(46) The second metal ring **145**, which serves as the outer ring of the second main bearing **126**, is located on the input side of the casing **122**. The second metal ring **145** is exposed on the input side of the casing **122**. The portion of the second metal ring **145** that protrudes in the axial direction beyond the depression of the casing **122** constitutes the inner sliding surface **148** serving as the outer ring. The inner sliding surface **148** is in contact with the second carrier (hold flange) **120**. The second metal ring **145** is firmly fitted onto the casing **122**, and the fit can be, for example, a clearance fit, an interference fit or a transition fit. The clearance may be determined by the difference in thermal expansion coefficient.

(47) The first main bearing **124** is arranged between the first carrier **118** and the casing **122**. The second main bearing **126** is arranged between the second carrier **120** and the casing **122**. The main bearings **124** and **126** of the present embodiment, shown in FIGS. **1** and **2**, are plain bearings. The main bearings **124** and **126** have the inner sliding surface **148** serving as part of the outer ring and an outer sliding surface **149** serving as part of the inner ring. The outer sliding surface **149** is formed on the outer peripheral surface of the carriers **118** and **120**. The inner sliding surface **148** is formed on the inner peripheral surface of the metal rings **144** and **145**.

(48) The inner sliding surface **148** serving as part of the outer ring of the main bearings **124** and **126** is made of a material having a higher thermal conductivity than the resin making the carriers **118** and **120** forming the outer sliding surface **149** serving as part of the inner ring.

(49) Any material can be used to make the metal rings **144** and **145** serving as the outer ring of the main bearings **124** and **126** as long as the material exhibits a higher thermal conductivity than the resin of the carriers **118** and **120** serving as the inner ring and a higher strength than the resin of the carriers **118** and **120**. For example, the material used to make the metal rings **144** and **145** may be metallic or non-metallic materials. The metal rings **144** and **145** relating to the present embodiment may be made of copper- or aluminum-based metals, or iron-based metals such as bearing steel.

(50) The metal rings **144** and **145** can be solid or hollow. The metal rings **144** and **145** may have a multi-layered structure with a core material being wrapped in a surface material forming the inner sliding surface **148**. For example, one of the core and surface materials of the multi-layered structure of the metal rings **144** and **145** may be made of iron-based metals and the other may be made of copper- or aluminum-based metals. As another example, the metal rings **144** and **145** may be made of sintered metals.

(51) As for the main bearings **124** and **126**, the inner and outer sliding surfaces **148** and **149** are outside the outer pins **117** in the radial direction. As shown in FIG. **1**, the first and second metal rings **144** and **145** both overlap the outer pins **117** when seen in the radial direction. This arrangement can contribute to maintain the strength of the casing **122** along the entire length.

(52) The inner pins **140** are inserted into the inner pin holes **141**, which are formed in the external gears **114**, with a clearance therebetween. The inner pins **140** are fitted at one end thereof into recesses **118a** in the first carrier **118** and at the other end thereof into recesses **120b** in the second

carrier **120**. The inner pins **140** are fixedly inserted into the recesses **118a** and **120b** with bolts **140a**. Alternatively, the inner pins **140** may be press-fitted into the recesses **118a** and **120b**. In this case, the inner pins **140** may not be necessarily fixed using bolts or the like. The inner pins **140** are in contact with part of the inner pin holes **141**, which are formed in the external gears **114**. With such arrangement, the inner pins **140** restricts the external gears **114** from rotating on their own axis and only allows the external gears **114** to oscillate. The inner pins **140** serve as a coupling member for transmitting power between the first and second carriers **118** and **120** and the external gears **114**.

(53) The carrier pins **138** are inserted into the carrier pin holes **139**, which are formed in the external gears **114**, with a clearance therebetween. The carrier pins **138** are fitted at one end thereof into recesses **118c** in the first carrier **118** and at the other end thereof into recesses **120c** in the second carrier **120**. The carrier pins **138** are fixedly inserted into the recesses **118c** and **120c** with bolts **138a**. Alternatively, the carrier pins **138** may be press-fitted into the recesses **118c** and **120c**. In this case, the carrier pins **138** may not be necessarily fixed using bolts or the like. The carrier pins **138** are not in contact with the carrier pin holes **139** formed in the external gears **114**. With such arrangement, the carrier pins **138** do not play role in restricting the external gears **114** from rotating on their own axis. The carrier pins **138** serve as a coupling member contributing only to couple together the first and second carriers **118** and **120**.

(54) Speed reducers have seen an increase in its applications, including collaborative robots working closely with people. In order to expand the applications, it is desired to reduce the weight and noise of the speed reducers. Conventional speed reducers are made up by components made of iron-based metals. In order to diminish the weight, the components may be made of materials with low specific gravity. Favorable materials with low specific gravity include resin. On the other hand, if the components are made of resin, the components may experience temperature rise due to poorer heat dissipation and resultantly have shortened lifetime. For this reason, the materials of the components are desirably selected in consideration of the purposes of weight reduction and heat dissipation. Here, the weight reduction may lead to compromised strength, which needs to be avoided.

(55) In the speed reducer **100**, a large amount of heat is often generated inside the speed reducer **100**, more specifically, around the main bearings **124** and **126**, which are plain bearings. In addition, a large amount of heat is also often generated around the input shaft **112**, which is rotatable at a relatively high speed. Furthermore, the speed reducer **100** may malfunction unless the inner pins **140** and external gears **114** maintain sufficient strength. As noted, the speed reducers experience a large temperature increase if the heat generated inside is dissipated outside less efficiently. The resin components experience a sudden drop in rigidity and strength as the temperature rises. Accordingly, the continued use of the less rigid and strong resin components may probably result in malfunction.

(56) To deal with this issue, one of the members that are movable relative to each other is made of a resin, the other member is desirably made of a material exhibiting a higher resistance against wear and a higher thermal conductivity [W/(m.Math.K)] than the resin member. In this case, the heat generated inside can be dissipated outside in a better manner than when the other member is made of a low-thermal-conductivity material. At the same time, the members can achieve a longer lifetime than when the other member is made of a low-wear-resistance material.

(57) The material used to make the inner sliding surface **148** serving as the outer ring may be made of any material as long as it exhibits higher wear-resistance and thermal conductivity than the resin used to make the internal gear body **116a** forming the outer sliding surface **149** serving as the inner ring. The inner sliding surface **148** can be made of, for example, metal, non-metal, and highly thermally conductive materials. The first and second metal rings **144** and **145**, which constitute the inner sliding surface **148** of the present embodiment, and the outer pins **117** may be made of iron- and aluminum-based metals such as bearing steel, light metals such as aluminum, magnesium,

beryllium, and titanium, or composites thereof. Alternatively, the first and second metal rings **144** and **145** and the outer pins **117** may be made of ceramics or other materials. As the other components such as the carriers **118** and **120** are made of resin, the speed reducer **100** can achieve both light weight and sufficient mechanical strength.

(58) The input shaft **112** and the input shaft bearings **134** located between the first carrier **118** and the input shaft **112** receive high-speed rotation before it is slowed down. This means that the input shaft **112** and the input shaft bearings **134** experience a relatively large temperature rise and allow the rotation to be input only at low speed if they exhibit low resistance against heat. For this reason, the input shaft bearings **134**, input shaft **112** and eccentric bearings **130** may be made of metals, such as iron-based metals. This can prevent a drop in the allowable number of rotations input into them. Since the input shaft **112** is subjected to large torsional stress, the input shaft **112** is preferably made of a more rigid material than the first carrier **118**. The input shaft **112** may be made of aluminum or iron-based metals with a higher torsional strength than aluminum, for example. The iron-based metals can include carbon steel, bearing steel, and stainless steel, any of which can be used depending on the desired characteristics.

(59) The carrier pins **138** are desirably highly rigid in order to reliably achieve sufficient coupling strength between the first and second carriers **118** and **120**. Considering this, the carrier pins **138** may be made of metals. According to the present example, the carrier pins **138** are made of a material that has higher wear-resistance and thermal conductivity than resin, for example, metals such as aluminum.

(60) The following now describes how the speed reducer **100** configured as described above operates. As rotational power is transmitted from the driving unit to the input shaft **112**, the eccentric portions **112a** of the input shaft **112** rotate around the rotational center line passing through the input shaft **112** (the central axis **1La**). This causes the eccentric portions **112a** to oscillate the external gears **114**. The external gears **114** oscillate in such a manner that their own axial center rotates around the rotational center line of the input shaft **112**. As the external gears **114** oscillate, the meshing between the external gears **114** and the outer pins **117** of the internal gear **116** takes place at sequentially shifted positions. Accordingly, every time the input shaft **112** rotates once, either the external gears **114** or internal gear **116** rotates on its own axis by an amount determined by the difference between the number of teeth of the external gears **114** and the number of outer pins **117** of the internal gear **116**. In the present embodiment, the rotation of the external gears **114** on their own axis results in reduced rotations being output through the first carrier **118** or casing **122**.

(61) In the speed reducer **100** of the present embodiment, the first and second main bearings **124** and **126** are constituted by the first and second metal rings **144** and **145**, and the first and second carriers **118** and **120**, which account for the main portion of the weight of the speed reducer **100**, are made of resin. This can contribute to successful weight reduction.

(62) In the speed reducer **100** of the present embodiment, the metal rings **144** and **145** forming the inner sliding surface **148** are made of a material exhibiting higher wear-resistance and thermal conductivity than the resin used to make the carriers **118** and **120** forming the outer sliding surface **149**. This can help dissipating the heat generated in the speed reducer **100** more efficiently via the metal rings **144** and **145** and the outer pins **117**. As a result, the present embodiment can prevent a temperature rise in the speed reducer **100** more effectively.

(63) In the speed reducer **100** of the present embodiment, as described above, the first and second metal rings **144** and **145** wrap the outer pins **117** when seen in the radial direction. This arrangement can contribute to maintain the strength of the casing **122** along the entire length in the axial direction. Therefore, the present embodiment can prevent deformation of the casing **122** and deformation of the carriers **118** and **120**, and also prevent failure of the external gears **114**. As a result, the present embodiment can maintain sufficient strength.

(64) In the present embodiment, the outer pins **117** are made of metals or the like. This can

contribute to prevent the failure of the external gears **114** and to maintain sufficient strength. As for the main bearings **124** and **126**, the inner and outer sliding surfaces **148** and **149** do not wrap the outer pins **117** when seen in the axial direction, and the inner and outer sliding surfaces **148** and **149** and the outer pins **117** are arranged substantially consecutively in the axial direction. With such arrangement, the thickness of the speed reducer **100** can be reduced while the speed reducer **100** maintains the strength. In other words, the dimension of the speed reducer **100** in the direction extending along the central axis **1La** can be reduced, so that the size reduction can be accomplished.

(65) The inner sliding surface **148** of the first and second metal rings **144** and **145** is close to the outer pins **117** in the radial direction. Accordingly, the heat transferred from the main bearings **124** and **126**, which are sliding bearings, to the first and second metal rings **144** and **145** can be dissipated outside, so that the speed reducer **100** can achieve better heat dissipation.

(66) If the inner and outer sliding surfaces **148** and **149** experience an excessive temperature rise, this may cause the resin on the surfaces to melt or the surfaces to stick together. In this case, the speed reducer **100** may fail. According to the present embodiment, however, improved heat dissipation can be achieved. This can prevent the heat from being contained in the speed reducer **100**, thereby preventing problems from occurring. In addition, since the outer pins **117** are made of metals or the like, the outer pins **117** can release the heat transferred from the vicinity of the main bearings **124** and **126**, to prevent a local temperature rise. As a result, the speed reducer **100** can achieve improved heat dissipation.

(67) At the same time, since the metal rings **144** and **145** and the carriers **118** and **120** constitute the main bearings **124** and **126**, metal or heavy components can be reduced in number when compared with the case where ball bearings, roller bearings, and other configurations are employed. In this manner, the speed reducer **100** can be reliably configured, while keeping sufficient heat dissipation and achieving further weight reduction.

Second Embodiment

(68) The following describes a speed reducer relating to a second embodiment of the disclosure with reference to the accompanying drawings. FIG. 3 is a sectional view showing the speed reducer relating to the second embodiment along a direction extending along a main axis. The present embodiment is different from the first embodiment in terms of the main bearings. The other components relating to the second embodiment have counterparts in the first embodiment, and are thus assigned with the same reference signs and not described now.

(69) In the speed reducer **100** of the present embodiment, as shown in FIG. 3, the first metal ring **144** of the first main bearing **124** is provided on the first carrier (shaft flange) **118**. The first metal ring **144** has the outer periphery constituting the outer sliding surface **149** of the main bearing **124**. The first carrier (shaft flange) **118** has a depression for receiving therein the first metal ring **144**. The first metal ring **144** is adjacent to the outer pins **117** in the axial direction. In the first carrier (shaft flange) **118**, the first metal ring **144** serves as the inner ring of the first main bearing **124**. The first metal ring **144** may have the outer periphery flush with the outer periphery of the first carrier (shaft flange) **118**.

(70) The first metal ring **144**, which serves as the inner ring of the first main bearing **124**, is located on the input side in the first carrier (shaft flange) **118**. The outer peripheral portion of the first metal ring **144**, which is flush with the outer periphery of the first carrier (shaft flange) **118**, forms the outer sliding surface **149** serving as the inner ring. The outer sliding surface **149** of the first metal ring **144** is in contact with the casing **122**. The first metal ring **144** is firmly fitted onto the first carrier (shaft flange) **118**, and the fit can be, for example, a clearance fit, an interference fit or a transition fit. The clearance may be determined by the difference in thermal expansion coefficient.

(71) The inner peripheral surface of the casing **122** in contact with the outer sliding surface **149** forms the inner sliding surface **148** serving as the outer ring of the first main bearing **124**.

Therefore, the outer ring of the first main bearing **124** is integrated with the casing **122**.

(72) Likewise, the second metal ring **145** of the second main bearing **126** is provided on the second carrier (hold flange) **120** in the speed reducer **100** of the present embodiment, as shown in FIG. 3. The second metal ring **145** has the outer periphery constituting the outer sliding surface **149** of the main bearing **126**.

(73) The second carrier (hold flange) **120** has a depression for receiving therein the second metal ring **145**. The second metal ring **145** is adjacent to the outer pins **117** in the axial direction. In the second carrier (hold flange) **120**, the second metal ring **145** serves as the inner ring of the second main bearing **126**. The second metal ring **145** may have the outer periphery flush with the outer periphery of the second carrier (hold flange) **120**.

(74) The second metal ring **145**, which serves as the inner ring of the second main bearing **126**, is located on the output side in the second carrier (hold flange) **120**. The outer peripheral portion of the second metal ring **145**, which is flush with the outer periphery of the second carrier (hold flange) **120**, forms the outer sliding surface **149** serving as the inner ring. The outer sliding surface **149** of the second metal ring **145** is in contact with the casing **122**. The second metal ring **145** is firmly fitted onto the second carrier (hold flange) **120**, and the fit can be, for example, a clearance fit, an interference fit or a transition fit. The clearance may be determined by the difference in thermal expansion coefficient.

(75) The inner peripheral surface of the casing **122** in contact with the outer sliding surface **149** forms the inner sliding surface **148** serving as the outer ring of the second main bearing **126**. Therefore, the outer ring of the second main bearing **126** is integrated with the casing **122**. Note that the metal rings **144** and **145** of the main bearings **124** and **126** can be alternatively configured to radially externally cover the entire outer periphery of the carriers **118** and **120**.

(76) In the speed reducer **100** of the present embodiment shown in FIG. 3, the inner and outer sliding surfaces **148** and **149** are respectively made of resin and metals. Accordingly, the second embodiment can produce the same effects as the first embodiment.

Third Embodiment

(77) The following describes a speed reducer relating to a third embodiment of the disclosure with reference to the accompanying drawings. FIG. 4 is a sectional view showing the speed reducer relating to the third embodiment along a direction extending along a main axis. FIG. 5 is a sectional view along a line V-V in FIG. 4. In FIGS. 4 and 5, the reference numeral **200** indicates the speed reducer.

(78) The speed reducer **200** relating to the third embodiment is an eccentric oscillation speed reducer having an external gear and an internal gear meshing with each other. Oscillating the external gear causes one of the internal and external gears to rotate on its own axis, so that the generated rotational component is output to a driven device through an output member.

(79) The speed reducer **200** of the present embodiment is different from the speed reducer **100** relating to the first embodiment in that two external gears **214** are assembled together with a phase difference of 180 degrees, as shown in FIGS. 4 and 5. Except for this, the speed reducer **200** relating to the third embodiment is the same as the speed reducer **100** relating to the first embodiment. The speed reducer **200** relating to the third embodiment includes an input shaft **212**, external gears **214**, an internal gear **216**, a first carrier **218**, a second carrier **220**, a casing **222**, a first main bearing **224**, a second main bearing **226**, inner pins **240** and carrier pins **238**. Hereinafter, the expression “main axis direction” refers to the direction extending along a central axis **2La** of the internal gear **216**, and the expressions “circumferential direction” and “radial direction” respectively refer to the circumferential and radial directions of the circle centered on the central axis **2La**. For convenience, the input side refers to one side in the axial direction (the right side in FIG. 4) and the non-input side refers to the other side (the left side in FIG. 4). In the following description, the first and second carriers **218** and **220** may be simply referred to as the carriers **218** and **220**. Likewise, the first and second main bearings **224** and **226** may be simply referred to as the main bearings **224** and **226**.

(80) The input shaft **212** is rotatable around the rotational center line when acted upon by rotational power input from a drive device (not shown). The speed reducer **200** relating to the third embodiment is a center crank type, where the rotational center line of the input shaft **212** is aligned with the central axis 2La of the internal gear **216**. The drive device is, for example, a motor, a gear motor or an engine.

(81) The input shaft **212** relating to the present embodiment is an eccentric shaft having a plurality of eccentric portions **212a** for oscillating the external gears **214**. Having such a configuration, the input shaft (eccentric body) **212** is sometimes referred to as a crankshaft. The axial center of the eccentric portions **212a** is eccentric to the rotational center line of the input shaft **212**. In the present embodiment, two eccentric portions **212a** are adjacent to each other. The adjacent eccentric portions **212a** are 180° out of phase in terms of eccentricity.

(82) The input shaft **212** is supported on the input side by a second cover **223** via an input shaft bearing **234** and on the non-input side by the first carrier **218** via an input shaft bearing **234**. In other words, the input shaft **212** is supported such that it is rotatable relative to the first carrier **218** and the second cover **223**. The input shaft bearings **234** are not limited to any particular configurations, but can be ball bearings with spherical rollers in the present embodiment. Preload may be applied to the input shaft bearings **234**, but no preload is applied in the present example.

(83) The internal gear **216** meshes with the external gears **214**. According to the present embodiment, the internal gear **216** includes an internal gear body **216a** integrated into the casing **222** and outer pins (internal tooth pins) **217** disposed in pin grooves **216b** formed in the internal gear body **216a** at intervals in the circumferential direction. The outer pins **217** are cylindrical pin members rotatably supported on the internal gear body **216a**. The outer pins **217** form the internal teeth of the internal gear **216**. The number of outer pins **217** (the number of internal teeth) of the internal gear **216** is slightly (in this example, by one) greater than the number of external teeth of the external gears **214**.

(84) The internal gear body **216a** is made of resin. The internal gear body **216a** can be made of a variety of resin. In the present embodiment, the internal gear body **216a** is made of, for example, polyacetal (POM). The present embodiment, however, is not limited to such, and the internal gear body **216a** may be made of other resin than POM, for example, polyaryletherketones (PAEK) including polyetheretherketone (PEEK).

(85) The resin used to make the internal gear body **116a** and other constituent parts of the present embodiment may be resin containing reinforcing fibers such as glass or carbon fibers, a resin without reinforcing fibers, a lamination of a base material such as paper or fabric impregnated with resin. The resin used for the constituent parts of the present embodiment may be a resin having thermally conductive fillers mixed therein. The thermally conductive fillers include, for example, nano-order fillers, powders of ceramics such as aluminum oxide and aluminum nitride, and powders of metals such as aluminum, copper and graphite.

(86) In the speed reducer **200**, a large amount of heat is often generated inside, in particular, around the input shaft **212**, which is rotatable at a relatively high speed. The speed reducer **200** experiences a large temperature rise if the heat generated inside is dissipated outside less efficiently. The resin components experience a sudden drop in strength as the temperature rises. Accordingly, the continued use of the resin components may probably result in breakage. To deal with this, one of the gears that mesh with each other is made of a resin, the other gear is desirably made of a material exhibiting a higher thermal conductivity [W/(m.Math.K)] than the resin gear. In the eccentric oscillation speed reducer **200**, the outer pins **217** are made of a material that has a higher thermal conductivity than the resin material used for the internal gear body **216a**. In this case, the heat generated inside can be dissipated outside in a better manner than when the outer pins **217** are made of a low thermal conductivity material.

(87) The outer pins **217** can be made of any material as long as it exhibits a higher thermal conductivity than the resin used to make the internal gear body **216a** and can be made of metal,

highly thermally conductive resin, non-metal materials and the like. Highly thermally conductive resin include resin having thermally conductive fillers mixed therein. The outer pins **217** may be, for example, made of resin having carbon nanotubes (CNTs), boron nitride nanotubes (BNNTs) mixed therein. The outer pins **217** relating to the present embodiment may be made of iron-based metals such as bearing steel.

(88) The outer pins **217** can be solid or hollow. The outer pins **217** may have a multi-layered structure with a core material being wrapped in a surface material. For example, one of the core and surface materials of the multi-layered structure of the outer pins **217** may be made of iron-based metals, and the other may be made of copper- or aluminum-based metals. Such a structure can eliminate the trade-off between the mechanical and thermal characteristics. As yet another example of the core and surface materials of the multi-layered structure of the outer pins **217**, one of them may be made of metals, and the other may be made of resin. In a further different example, the outer pins **117** may be made of sintered metals.

(89) The external gears **214** are provided in a one-to-one correspondence with the eccentric portions **212a**. The external gears **214** are rotatably supported on the eccentric portions **212a** via eccentric bearings **230**. The external gears **214** have twelve through holes arranged at equal intervals and offset from the axis of the external gears **214**. Three of the twelve through holes, which are at equal intervals of 120 degrees around the axis of the external gears **214**, serve as carrier pin holes **239**. The carrier pin holes **239** receive carrier pins **238**. The remaining nine through holes serve as inner pin holes **241**. The inner pin holes **241** receive inner pins **240**. In the present embodiment, the diameter of the carrier pin holes **239** is greater than that of the inner pin holes **241**. The present embodiment, however, is not limited to such, and the carrier pin holes **239** may have the same diameter as the inner pin holes **241**.

(90) The external gears **214** are made of resin. The external gears **214** can be made of a variety of types of resin. In particular, since the external gears **214** are located near the input shaft **212**, which experiences a large temperature rise, the external gears **214** may be made of resin that have a higher heat resistance temperature than the internal gear body **216a**. Considering this, the external gears **214** are made of, for example, PEEK. The present embodiment, however, is not limited to such, and the external gears **214** may be made of, for example, POM or other resin than PEEK.

(91) The carrier pin holes **239** and the inner pin holes **241** are at the same distance from the center in the radial direction and shaped like a circle. Corrugated teeth are formed on the outer periphery of the external gears **214**. The external gears **214** oscillate within a plane to which the axis of the external gears **214** is normal, such that the corrugated teeth move while keeping touching the internal gear **216**. A gap is provided between the inner pins **240** and the inner pin holes **241** to absorb the oscillating component of the external gears **214**. The inner pins **240** partially touch the inner wall surface of the inner pin holes **241**.

(92) The carriers **218** and **220** are arranged on the lateral portion of the external gears **214** in the axial direction. The first carrier (shaft flange) **218** is located on the non-input-side lateral portion of the external gears **214**. The second carrier (hold flange) **220** is located on the input-side lateral portion of the external gears **214**. The first and second carriers **218** and **220** are rotatably supported by the casing **222** via first and second main bearings **224** and **226**. The carriers **218** and **220** are shaped like a disc as a whole. The first carrier **218** rotatably supports the input shaft **212** via an input shaft bearing **234**. The second carrier **220** may be configured to support the input shaft **212** via an input shaft bearing **234**, but does not support the input shaft bearing **234** and the input shaft **212** in the present embodiment.

(93) The first and second carriers **218** and **220** are coupled to each other via the carrier and inner pins **238** and **240**. The carrier and inner pins **238** and **240** are positioned offset from the axial center of the external gears **214** in the radial direction and penetrate the external gears **214** in the axial direction. In the present embodiment, the carrier and inner pins **238** and **240** are provided as separate parts from the carriers **218** and **220**. The present embodiment, however, is not limited to

such. As another example, the carrier and inner pins **238** and **240** may be partly integrated into the carriers **218** and **220**.

(94) One of the first carrier **218** and the casing **222** serves as an output member for outputting rotational power to a driven device, and the other serves as a stationary member fixedly attached to an external member for supporting the speed reducer **200**. The output member is rotatably supported on the stationary member via the main bearings **224** and **226**. In the present embodiment, the output member is the first carrier **218**, and the stationary member is the casing **222**. The non-input-side end surface of the first carrier **218** may have a driven member **250**, which is to be rotated by the speed reducer **200**, coupled thereto with bolts **250b**. The bolts **250b** relating to the present embodiment may be made of iron-based metals.

(95) The casing **222** is shaped like a hollow tube as a whole. The internal gear **216** is provided on the inner periphery of the casing **222**. A flange or the like may be provided on the outer periphery of the casing **222**, but no flange is provided in the present example. The casing **222** has a first cover **221** covering the casing **222** on the non-input side and a second cover **223** covering the casing **222** on the input side. The first and second covers **221** and **223** are fixedly attached to the casing **222** with a plurality of bolts arranged next to each other in the circumferential direction.

(96) The casing **222** has a depression for receiving therein the input side of the outer ring of the first main bearing **224**. The first cover **221** has a depression for receiving therein part of the non-input side of the outer ring of the first main bearing **224**. The outer ring of the first main bearing **224** is supported by and sandwiched between the casing **222** and the first cover **221** in the axial direction. The casing **222** has a depression for receiving therein the non-input side of the outer ring of the second main bearing **226**. The second cover **223** has a depression for receiving therein part of the input side of the outer ring of the second main bearing **226**. The outer ring of the second main bearing **226** is supported by and sandwiched between the casing **222** and the second cover **223** in the axial direction. The second cover **223** has another depression for receiving therein the outer ring of the input-side input shaft bearing **234**. In other words, the second cover **223** rotatably supports the input shaft **212** on the input side thereof via the input shaft bearing **234**.

(97) The first main bearing **224** is arranged between the first carrier **218** and the casing **222**. The second main bearing **226** is arranged between the second carrier **220** and the casing **222**. The main bearings **224** and **226** relating to the present embodiment respectively have metal rings **244** and **245** forming the inner sliding surface **248**. The metal rings **244** and **245** rotatably supports the carriers **218** and **220**.

(98) The main bearings **224** and **226** are plain bearings. The main bearings **224** and **226** have (i) the metal rings **244** and **245** forming the inner sliding surface **248** serving as the outer ring and (ii) an outer sliding surface **249** serving as the inner ring. The outer sliding surface **249** is formed on the outer peripheral surface of the carriers **218** and **220**. The metal rings **244** and **245** serving as the outer ring are firmly fitted onto the casing **222**, and the fit can be, for example, a clearance fit, an interference fit or a transition fit. The clearance may be determined by the difference in thermal expansion coefficient. Pressure may be applied to the main bearings **224** and **226**, but no pressure is applied in the present example.

(99) In the present embodiment, the metal rings **244** and **245** serving as the outer ring of the main bearings **224** and **226** are made of a material having a higher thermal conductivity than the resin used for the internal gear body **216a**. Furthermore, the metal rings **244** and **245** are in contact with the outer pins **217** in the axial direction. As shown in FIG. 4, the end of the metal rings **244** and **245** may be in direct contact with the end of the outer pins **217**. The present embodiment, however, is not limited to such, and the end of the metal rings **244** and **245** may be in contact with the end of the outer pins **217** with a spacer being provided therebetween in the axial direction. The spacer may be made of a material exhibiting a higher thermal conductivity than the resin used for the internal gear body **216a**. With such arrangement, the heat transferred to the outer pins **217** is dissipated to the carriers **218** and **220** and the casing **222** via the metal rings **244** and **245**. This can contribute to

achieve improved heat dissipation. In addition, the heat transferred to the first carrier **218** is dissipated to the outside through the driven member **250**. This can contribute to achieve improved heat dissipation.

(100) The metal rings **244** and **245** and the outer pins **217** can be made of any material as long as it exhibits higher thermal conductivity than the resin used to make the internal gear body **216a** and can be made of metal, highly thermally conductive resin, non-metal materials and the like. The metal rings **244** and **245** and the outer pins **217** relating to the present embodiment may be made of iron-based metals such as bearing steel.

(101) The first carrier **218** can be made of any material as long as it exhibits higher thermal conductivity than the resin used to make the internal gear body **216a** and can be made of metal, highly thermally conductive resin, non-metal materials and the like. In order to achieve both light weight and sufficient mechanical strength, the first carrier **218** may be made of light metals (metals with a specific gravity of 4, 5 or less) such as aluminum, magnesium, beryllium and titanium or a combination of these. The first carrier **218** relating to the present embodiment is made of aluminum-based metals. In this way, the first carrier **218** can be made of a metal material with a lower specific gravity than is the input shaft **212**.

(102) The second carrier **220** can be made of metals or various types of resin. In the present embodiment, the second carrier **220** is made of POM, for example. In this way, the second carrier **220** can achieve a reduced weight. In order to diminish the heat conduction from the input shaft bearing **234**, the second carrier **220** is not in direct contact with the input shaft bearing **234** but arranged with a space being provided therebetween. The second carrier **220** may be made of a material having a higher thermal conductivity than the resin used for the internal gear body **216a**. This can contribute to further improve the heat dissipation.

(103) The inner pins **240** are inserted into the inner pin holes **241**, which are formed in and penetrate the external gears **214**, with a gap therebetween. The inner pins **240** are fitted at one end thereof into recesses **218a** in the first carrier **218** and at the other end thereof into recesses **220b** in the second carrier **220**. The inner pins **240** are press-fitted in the recesses **218a** and **220b** and not secured using bolts or the like. The inner pins **240** are in contact with part of the inner pin holes **241**, which are formed in the external gears **214**. With such arrangement, the inner pins **240** restricts the external gears **214** from rotating on their own axis and only allows the external gears **114** to oscillate. The inner pins **240** serve as a coupling member playing a role in transmitting power between the first and second carriers **218** and **220** and the external gears **214**.

(104) The carrier pins **238** are inserted into the carrier pin holes **239**, which are formed in the external gears **214**, with a gap therebetween. The carrier pins **238** are fitted at one end thereof into recesses **218c** in the first carrier **218** and at the other end thereof into recesses **220c** in the second carrier **220**. The carrier pins **238** are press-fitted in the recesses **218c** and **220c** and not secured using bolts or the like. The carrier pins **238** are surrounded by tubular spacers **237**. The spacers **237** touch, at one end thereof, the first carrier **218** and, at the other end thereof, the second carrier **220**. The spacers **237** are designed to keep an appropriate distance in the axial direction between the first carrier **218** and the second carrier **220**. The carrier pins **238** and spacers **237** are not in contact with the carrier pin holes **239** formed in the external gears **214**, so that they do not contribute to restrain the external gears **214** from rotating on their own axis. The carrier pins **238** serve as a coupling member contributing only to couple together the first and second carriers **218** and **220**.

(105) The constituents of the speed reducer relating to the present embodiment are desirably made of a material selected in consideration of the purposes of weight reduction and heat dissipation. In recent years, speed reducers have seen an increase in its applications including collaborative robots working closely with people. For this reason, it is desired to reduce the weight and noise of the speed reducers. Conventional speed reducers are constituted by parts made of iron-based metals. In order to reduce the weight, the constituents may be made of materials with low specific gravity. Suitable materials with low specific gravity include resin. On the other hand, if the components are

made of resin, they may experience a temperature rise due to poorer heat dissipation and resultantly have shortened lifetime.

(106) The input shaft **212** and the input shaft bearings **234** located between the first carrier **218** and the input shaft **212** receive high-speed rotation input thereto before it is slowed down. This means that the input shaft **212** and the input shaft bearings **234** experience a relatively large temperature rise and they allow a smaller number of rotations input thereto if they exhibit low resistance against heat. For this reason, the input shaft bearings **234**, input shaft **212** and eccentric bearings **230** may be made of metals, such as iron-based metals. This can prevent a drop in the allowable number of rotations input into them. Since the input shaft **212** is subjected to large torsional stress, the input shaft **212** is preferably made of a more rigid material than the first carrier **218**. The input shaft **212** is made of iron-based metals with a higher torsional strength than aluminum. The iron-based metals that can be used to make the constituents of the speed reducer relating to the present embodiment can include carbon steel, bearing steel, and stainless steel, any of which can be used depending on the desired characteristics.

(107) The carrier pins **238** are desirably highly rigid in order to reliably achieve sufficient coupling strength between the first and second carriers **218** and **220**. Considering this, the carrier pins **238** may be made of metals and the spacers **237** may be made of resin in order to achieve reduced weight. In the present embodiment, the carrier pins **238** are made of iron-based metals and the spacers **237** are made of POM. The casing **222** is integrated with the internal gear body **216a** and may be made of the same material as the internal gear body **216a**. For the purposes of weight reduction, the first and second covers **221** and **223** may be made of resin. The first and second covers **221** and **223** may be made of the same resin or different types of resin. In the present embodiment, the first and second covers **221** and **223** may be made of POM, for example.

(108) The following now describes how the speed reducer **200** works. As rotational power is transmitted from the driving unit to the input shaft **212**, the eccentric portions **212a** of the input shaft **212** rotate around the rotational center line of the input shaft **212**. This causes the eccentric portions **212a** to oscillate the external gears **214**. The external gears **214** oscillate in such a manner that their own axial center rotates around the rotational center line of the input shaft **212**. As the external gears **214** oscillate, the meshing between the external gears **214** and the outer pins **217** of the internal gear **216** takes place at sequentially shifted positions. Accordingly, every time the input shaft **212** rotates once, either the external gears **214** or the internal gear **216** rotates on its own axis by an amount determined by the difference between the number of teeth of the external gears **214** and the number of outer pins **217** of the internal gear **216**. In the present embodiment, the external gears **214** rotate on their own axis, which results in reduced rotations being output through the first carrier **118**.

(109) In the speed reducer **200** of the present embodiment, the main bearings **224** and **226** have no metal constituents heavier than resins, except for the metal rings **244** and **245**. Thus, the speed reducer **200** can achieve further weight reduction.

(110) The speed reducer **200** relating to the present embodiment can produce the same effects as the speed reducers relating to the above-described embodiments.

Fourth Embodiment

(111) The following describes a speed reducer relating to a fourth embodiment of the disclosure with reference to the accompanying drawings. FIG. **6** is a sectional view showing the speed reducer relating to the fourth embodiment along the axial direction. The present embodiment is different from the first embodiment in terms of the main bearings. The other components relating to the fourth embodiment have counterparts in the first embodiment, and they are assigned with the corresponding reference signs with the digit in the hundreds place is 3 in place of 1, and not described now.

(112) In a speed reducer **300** relating to the present embodiment, as shown in FIG. **6**, a casing **322** has an inner sliding surface **348** integrated therewith, serving as the outer ring of first and second

main bearings **324** and **326**. The casing **322** is entirely made of a material exhibiting higher wear-resistance and higher thermal conductivity than the resin used to make an outer sliding surface **349** serving as the inner ring of the first and second main bearings **324** and **326**. Specifically, the casing **322** is made of metals. In the following description, the first and second main bearings **324** and **326** may be simply referred to as the main bearings **324** and **326**.

(113) In the speed reducer **300** relating to the present embodiment, as shown in FIG. **6**, the inner and outer sliding surfaces **348** and **349** are shaped such that they have a larger diameter at a position away from external gears **314** than at a position close to the external gears **314** in the direction extending along the central axis **3La**. The inner and outer sliding surfaces **348** and **349** form an angle of, for example, 30° to 60° relative to the central axis **3La**. The present embodiment is, however, not limited to such, and the inner and outer sliding surfaces **348** and **349** may form an angle in the range of 40° to 50° , more preferably 45° relative to the central axis **3La**.

(114) Alternatively, in the speed reducer **300** relating to the present embodiment, the inner and outer sliding surfaces **348** and **349** form an angle of, for example, 45° to 40° relative to the central axis **3La**. In this case, only a small angle is made by the end surface (the outer peripheral surface) of first and second carriers (shaft flange) **318** and (hold flange) **320**, which are made of resin. This can save the first and second carriers **318** and **320** from deformation. Stated differently, the first carrier (shaft flange) **318** and the second carrier (hold flange) **320**, which are made of resin, can be prevented from becoming too thin in the axial direction near their outer edge.

(115) In the speed reducer **300** relating to the present embodiment, the casing **322** and the first carrier (shaft flange) **318** tightly seal the internal space of the speed reducer **300** via the first main bearing **324**. Likewise, the casing **322** and the second carrier (hold flange) **320** tightly seal the internal space of the speed reducer **300** via the second main bearing **326**. The external gears **314** and the outer pins (internal tooth pins) **317** are housed in the internal space of the speed reducer **300**. The main bearings **324** and **326** are plain bearings. The internal space of the speed reducer **300** is tightly sealed by the inner and outer sliding surfaces **348** and **349**.

(116) In the speed reducer **300** relating to the present embodiment, a drive gear **313**, which is configured to input a driving force to the input shaft (eccentric body) **312**, is provided and rotatable integrally with a drive shaft **313a**. FIG. **6** does not show the carrier pins.

(117) In the speed reducer **300** relating to the present embodiment, the casing **322** is entirely made of a thermally conductive material that exhibits a higher wear resistance than resin, for example, metals. This allows the heat generated by the main bearings **324** and **326** to be quickly dissipated outside through the casing **322**. As a result, the temperature rise of the speed reducer **300** can be effectively prevented. In addition, since the constituents other than the casing **322**, the inner pins **340** and the drive gear **313** are made of resin, the speed reducer **300** can achieve reduced weight. Furthermore, since the casing **322** is made of metals, the speed reducer **300** can maintain sufficient strength and rigidity.

(118) In the speed reducer **300** relating to the present embodiment, the inner and outer sliding surfaces **348** and **349** form an angle in the above-described range relative to the central axis **3La**. This means that the inner and outer sliding surfaces **348** and **349** can have an increased area without causing an increase in the thickness of the speed reducer **300** in the axial direction. Furthermore, since the present embodiment can prevent deformation of the casing **322**, the first carrier (shaft flange) **318** and the second carrier (hold flange) **320**, the speed reducer **300** can be strong enough to prevent any malfunctions. As a result, the speed reducer **300** can avoid failures or the like and can stably keep operating.

(119) Furthermore, the speed reducer **300** relating to the present embodiment can produce the same effects as the speed reducers relating to the above-described embodiments.

Fifth Embodiment

(120) The following describes a speed reducer relating to a fifth embodiment of the disclosure with reference to the accompanying drawings. FIG. **7** is an enlarged sectional view showing a portion of

the speed reducer relating to the fifth embodiment that is near the main bearings, obtained by cutting the speed reducer along the axial direction. The present embodiment is different from the fourth embodiment described above in terms of the main bearings. Except for this difference, the present embodiment includes the same constituents as in the fourth embodiment, which are denoted by the same reference numerals and are not described here.

(121) In the speed reducer **300** relating to the present embodiment, as shown in FIG. 7, the inner and outer sliding surfaces **348** and **349** of the main bearings **324** and **326** bulge out or project in the direction extending along the central axis **3La**. Alternatively, one of the inner and outer sliding surfaces **348** and **349** may bulge out or project in the direction along the central axis **3La**.

(122) The inner and outer sliding surfaces **348** and **349** preferably touch each other along the entire surface, so that they slidably support each other with touching each other by a large area. This ideal condition, however, may encounter difficulties due to distortions caused by the manufacturing process. This can result in backlash, which may cause the speed reducer **300** to fail. To prevent the speed reducer **300** from failing, the inner and outer sliding surfaces **348** and **349** include crush allowance as precautions.

(123) The crush allowance of the inner and outer sliding surfaces **348** and **349** can be a projection having a curved cross-section, as shown in FIG. 7. With such configuration, the inner and outer sliding surfaces **348** and **349** contact each other over a range determined by the crush allowance. Accordingly, the range over which they touch each other can be reliably acquired. This can prevent manufacturing errors from keeping the inner and outer sliding surfaces **348** and **349** away from each other, as a result of which the inner and outer sliding surfaces **348** and **349** can reliably touch each other at the middle or in the vicinity thereof in the axial direction. Accordingly, the first carrier (shaft flange) **318** and the second carrier (hold flange) **320** can operate in a more stable manner relative to the casing **322**. Furthermore, since the present embodiment can prevent deformation of the casing **322**, the first carrier (shaft flange) **318** and the second carrier (hold flange) **320**, the speed reducer **300** can be strong enough to prevent any malfunctions. In addition, the temperature rise experienced by the main bearings **324** and **326** can be also prevented.

(124) Furthermore, the speed reducer **300** relating to the present embodiment can produce the same effects as the speed reducers relating to the above-described embodiments.

Sixth Embodiment

(125) The following describes a speed reducer relating to a sixth embodiment of the disclosure with reference to the accompanying drawings. FIG. 8 is a sectional view showing a casing constituting an inner sliding surface of a main bearing of the speed reducer relating to the sixth embodiment of the present disclosure. The present embodiment is different from the fourth embodiment described above in terms of the main bearings. Except for this difference, the present embodiment includes the same constituents as in the fourth embodiment, which are denoted by the same reference numerals and are not described here.

(126) In the speed reducer **300** relating to the present embodiment, as shown in FIG. 8, a casing **322** forms an inner sliding surface **348** with grooves **360** formed therein. The grooves **360**, which are formed in the inner sliding surface **348**, include radial grooves (groove) **361** extending in the radial direction and circumferential grooves (grooves) **362** and **363** extending in the circumferential direction. The radial grooves **361** are formed on the inner sliding surface **348** and spaced away from each other in the circumferential direction. The radial grooves **361** may be arranged at equal intervals in the circumferential direction, for example.

(127) The circumferential grooves **362** are near the middle of the inner sliding surface **348** in the axial direction. The circumferential grooves **363** are formed in the inner sliding surface **348** near the external gear **314** in the axial direction. The radial grooves **361** are not closer to the external gears **314** than are the circumferential grooves **363** in the axial direction. The radial grooves **361** extend to the edge of the inner sliding surface **348** in a direction away from the external gear **314** in the axial direction, to become in communication with the outside.

(128) The circumferential grooves **362** and **363** continuously extend in the circumferential direction of the inner sliding surface **348**, making a full circle. The radial grooves **361** are connected to the circumferential grooves **363** such that they terminate in the circumferential grooves **363**. In the inner sliding surface **348**, no other grooves are formed between the input shaft **312** and the circumferential grooves **363**. This means that the grooves **360** are separated from the outer pins **317** and not in contact with the outer pins **317**. The grooves **360** may be all equal in depth and width, or the circumferential grooves **362** and **363** may be larger than the radial grooves **361**.

(129) According to the speed reducer **300** relating to the present embodiment shown in FIG. **8**, the grooves **360** can catch any foreign matter such as dust and particles that enter between the inner sliding surface **348** and the outer sliding surface **349**. This can prevent foreign matters from entering the inner space within the speed reducer **300**. In other words, the present embodiment can prevent foreign substances such as dust and particles from affecting how the external gears **314**, outer pins **317** and the like work. In addition, the grooves **360** can catch excessive grease, lubricant or the like between the inner sliding surface **348** and the outer sliding surface **349**.

(130) Furthermore, the speed reducer **300** relating to the present embodiment can produce the same effects as the speed reducers relating to the above-described embodiments.

(131) Furthermore, the features of the present embodiment shown in FIG. **8** can be combined with, for example, the features of the fifth embodiment shown in FIG. **7**. FIG. **9** is an enlarged sectional view showing the casing **322** to indicate a relation between the grooves **360** and the crush allowance of the inner sliding surface **348** in the speed reducer **300** relating to the sixth embodiment. FIG. **10** is a sectional view showing the casing **322** to indicate another example of the relation between the grooves **360** and the crush allowance of the inner sliding surface **348** in the speed reducer **300** relating to the sixth embodiment. Specifically, a plurality of projections to serve as crush allowance having an arc-shaped cross-section are formed on the inner sliding surface **348**, and grooves can be formed between the projections.

(132) For example, as shown in FIG. **9**, the casing **322** has two parallel ridges running in the circumferential direction and having an arc-shaped cross-section, and the space between the circumferential protrusions can serve as the circumferential grooves **363**. The portion of the inner sliding surface **348** that is radially close to the input shaft **312** has curved ridges **348a**. In addition, the portion of the inner sliding surface **348** that is radially close to the outer peripheral edge of the casing **322** has curved ridges **348b**. The ridges **348a** and **348b** are formed such that they have substantially the same curvature in the cross-section. Between the ridges **348a** and **348b**, the circumferential grooves **363** are formed. In this way, a crush allowance can be provided on the inner sliding surface **348**, which improves the stability of operation and prevents contamination inside the speed reducer **300**.

(133) Alternatively, as shown in FIG. **10**, the curvature of the ridges **348a** in the cross-section can be greater than that of the ridges **348b** in the cross-section. In this way, the circumferential grooves **363** can be formed at a position further closer to the input shaft **312** in the radial direction.

(134) In the sixth embodiment shown in FIGS. **8** to **10**, the grooves **360** are formed on the inner sliding surface **348** of the casing **322**. The grooves, however, can be formed on the outer sliding surface **349**, for example. In this case, the outer sliding surface **349** is preferably made of metals, instead of resin. This means that the inner sliding surface **348** can be made of resin. In other words, as in the second embodiment shown in FIG. **3**, the carriers **318** and **320** can have metal rings provided thereon, and the outer sliding surface **349** can have grooves formed therein.

Seventh Embodiment

(135) The following describes a speed reducer relating to a seventh embodiment of the disclosure with reference to the accompanying drawings. FIG. **11** is a sectional view showing the speed reducer relating to the seventh embodiment along the axial direction. FIG. **12** is a sectional view along the line XII-XII in FIG. **11**. In FIGS. **11** and **12**, the reference numeral **400** indicates the speed reducer.

(136) The eccentric oscillation speed reducer **400** relating to the present embodiment is applied as a speed reducer to, for example, swivels of robots such as revolving drums and arm joints and swivels of various types of machine tools such as collaborative robots. The speed reducer **400** rotates, for example, at speeds from 80 rpm to 200 rpm, when used.

(137) In the speed reducer **400** of the present embodiment, as shown in FIGS. **11** and **12**, as an input shaft **408** is rotated, a crankshaft (eccentric body) **410** is rotated. In response to the rotation of eccentric portions **410a** and **410b** of the crankshaft **410**, the first and second external gears **414** and **416** are oscillatorily rotated. In this way, the input rotation is slowed down, and the reduced rotations can be output.

(138) The speed reducer **400** includes a casing (outer tube) **422**, a carrier **404**, an input shaft **408**, a plurality of (for example, three) crankshafts **410**, a first external gear **414**, a second external gear **416** and a plurality of (for example, three) transmission gears **420**.

(139) The casing **422** forms the outer surface of the speed reducer **400** and has a substantially cylindrical shape. A large number of pin grooves **422b** are formed on the inner peripheral surface of the casing **422**. Each pin groove **422b** extends in the axial direction of the casing **422** and has a semicircular sectional shape when cut along the plane orthogonal to the axial direction. The pin grooves **422b** are arranged at regular intervals and next to each other in the circumferential direction on the inner peripheral surface of the casing **422**.

(140) The casing **422** has a large number of internal tooth pins (outer pins) **417**. Each internal tooth pin **417** is attached in a corresponding one of the pin grooves **422b**. More specifically, each internal tooth pin **417** is fitted in the corresponding pin groove **422b** and retained therein such that it extends in the axial direction of the casing **422**. In this manner, the internal tooth pins **417** are at equal intervals in the circumferential direction of the casing **422**. The internal tooth pins **417** mesh with first external teeth **414a** of the first external gear **414** and second external teeth **416a** of the second external gear **416**. The internal tooth pins **417** form an internal gear **417A**.

(141) The carrier **404** is housed within the casing **422** while being aligned with the casing **422**. The carrier **404** is rotatable relative to the casing **422** about the same axis. Specifically, the carrier **404** is positioned inside the casing **422** in the radial direction. The carrier **404** is supported by first and second main bearings **424** and **426**, which are spaced away from each other in the axial direction, on the casing **422** such that the carrier **404** and the casing **422** are rotatable relative to each other. In the following description, the first and second main bearings **424** and **426** may be simply referred to as the main bearings **424** and **426**.

(142) The carrier **404** includes a base and a second carrier (hold flange) **404b**. The base has a first carrier (shaft flange) **404a** and a plurality of (for example, three) shaft portions **404c**.

(143) The first carrier **404a** is positioned near one of the ends of the casing **422** in the axial direction. The first carrier **404a** has a through hole **404d** formed at the radial center thereof. Around the through hole **404d**, a plurality of (for example, three) crankshaft mounting holes **404e** (hereinafter referred to simply as “mounting holes **404e**”) are arranged next to each other at equal intervals in the circumferential direction.

(144) The second carrier **404b** is positioned away from the first carrier **404a** in the axial direction and near the other of the ends of the casing **422** in the axial direction. The second carrier **404b** has a through hole **404f** formed at the radial center thereof. Around the through hole **404f**, a plurality of (for example, three) crankshaft mounting holes **404g** (hereinafter referred to simply as “mounting holes **404g**”) are provided. The mounting holes **404g** are positioned correspondingly to the mounting holes **404e** formed in the first carrier **404a**. Inside the casing **422**, a closed space (internal space) is defined by the inner surfaces of the second and first carriers **404b** and **404a** and the inner peripheral surface of the casing **422**.

(145) The three shaft portions **404c** are integrated with the first carrier **404a**. The shaft portions **404c** extend linearly from one of the main surfaces (the inner main surface) of the first carrier **404a** toward the second carrier **404b**. The three shaft portions **404c** are arranged next to each other at

equal intervals in the circumferential direction (see FIG. 12). The shaft portions **404c** are fastened to the second carrier **404b** with bolts **404h** (see FIG. 11). In this manner, the first carrier **404a**, the shaft portions **404c** and the second carrier **404b** are integrated with each other.

(146) The first and second carriers **404a** (shaft flange) and **404b** (hold flange) are rotatably supported by the casing **422** via the first and second main bearings **424** and **426**. The first carrier **404a** is rotatably supported by the casing **422** via the first main bearing **424**. The second carrier **404b** is rotatably supported by the casing **422** via the second main bearing **426**. The casing **422** has a first metal ring **444** located on the first carrier **404a** side in the axial direction of the input shaft **408** and a second metal ring **445** located on the second carrier **404b** side in the axial direction of the input shaft **408**. The first and second metal rings **444** and **445** are integrated into the casing **422**.

(147) The casing **422** has a depression for housing therein the first metal ring **444**. The first metal ring **444** is housed within the depression formed in the casing **422** in the axial direction. The first metal ring **444** serves as the outer ring of the first main bearing **424**. The outer periphery of the first metal ring **444** is connected to the inner periphery of the casing **422**.

(148) The first metal ring **444**, which serves as the outer ring of the first main bearing **424**, is located on the front end side of the input shaft **408** in the axial direction. The inner peripheral surface of the first metal ring **444** constitutes an inner sliding surface **448** serving as an outer ring. The inner sliding surface **448** is in contact with the outer peripheral surface of the first carrier **404a**. The first metal ring **444** is firmly fitted onto the casing **422**, and the fit can be, for example, a clearance fit, an interference fit or a transition fit. The clearance may be determined by the difference in thermal expansion coefficient.

(149) The casing **422** has a depression for housing therein the second metal ring **445**. The second metal ring **445** is housed within the depression formed in the casing **422** in the axial direction. The second metal ring **445** serves as the outer ring of the second main bearing **426**. The outer periphery of the second metal ring **445** is connected to the inner periphery of the casing **422**.

(150) The second metal ring **445**, which serves as the outer ring of the second main bearing **426**, is located on the base end side of the input shaft **408** in the axial direction. The inner peripheral surface of the second metal ring **445** constitutes an inner sliding surface **448** serving as the outer ring. The inner sliding surface **448** is in contact with the outer peripheral surface of the second carrier **404b**. The second metal ring **445** is firmly fitted onto the casing **422**, and the fit can be, for example, a clearance fit, an interference fit or a transition fit. The clearance may be determined by the difference in thermal expansion coefficient.

(151) The first main bearing **424** is sandwiched between the first carrier **404a** and the casing **422**. The second main bearing **426** is sandwiched between the second carrier **404b** and the casing **422**. The main bearings **424** and **426** of the present embodiment are plain bearings. The main bearings **424** and **426** have the inner sliding surface **448** serving as the outer ring and an outer sliding surface **449** serving as the inner ring. The outer sliding surface **449** is formed on the outer peripheral surface of the first and second carriers **404a** and **404b**.

(152) The inner sliding surface **448**, which is formed on the first and second metal rings **444** and **445**, and the outer sliding surface **449** are positioned at substantially the same distance in the radial direction as are the internal tooth pins **417**. As shown in FIGS. 11 and 12, the first and second metal rings **444** and **445** are both in contact with the internal tooth pins **417** in the direction extending along the central axis (main axis) **4La** of the input shaft **408**. Since the first and second metal rings **444** and **445** and the internal tooth pins **417** are in contact with each other in the axial direction, the speed reducer **400** can keep the strength along the entire length of the casing **422** in the axial direction.

(153) The input shaft **408** serves as an input part to which a driving force of a driving motor (not shown) is input. The input shaft **408** is inserted through the through hole **404f** in the second carrier **404b** and the through hole **404d** in the first carrier **404a**. The input shaft **408** is disposed such that the central axis **4La** is aligned with the axis of the casing **422** and the carrier **404** and rotatable

around the axis. An input gear **408a** is provided on the outer peripheral surface of the front end of the input shaft **408**.

(154) The three crankshafts **410** are arranged, within the casing **422**, at regular intervals around the input shaft **408** (see FIG. 12). The crankshafts **410** are each supported by a pair of crank bearings **412a** and **412b** such that they are rotatable about an axis relative to the carrier **404** (see FIG. 11). More specifically, a first crank bearing **412a** is mounted on a portion of each crankshaft **410** that is at a predetermined distance from one of the ends of the crankshaft **410** in the axial direction inwardly in the axial direction. The first crank bearings **412a** are mounted in the mounting holes **404e** in the first carrier **404a**. A second crank bearing **412b** is attached to the other end of each crankshaft **410** in the axial direction. The second crank bearings **412b** are mounted in the mounting holes **404g** in the second carrier **404b**. In this manner, the crankshafts **410** are rotatably supported by the first and second carriers **404a** and **404b**.

(155) Each crankshaft **410** has a shaft body **412c** and eccentric portions **410a** and **410b** integrated with the shaft body **412c**. The first and second eccentric portions **410a** and **410b** are arranged next to each other in the axial direction between the crank bearings **412a** and **412b**. The first and second eccentric portions **410a** and **410b** are each shaped like a circular cylinder. The first and second eccentric portions **410a** and **410b** project radially outward from the shaft body **412c** as they are arranged eccentrically to the axis of the shaft body **412c**. The first and second eccentric portions **410a** and **410b** are arranged with predetermined eccentricities from the axis and have a phase difference of a predetermined angle from each other.

(156) One of the ends of each crankshaft **410**, specifically, the portion mounted in the mounting hole **404e** in the first carrier **404a** has a mating portion **410c** to which the transmission gear **420** is mounted.

(157) The first external gear **414** is located in the closed space within the casing **422** and is attached to the first eccentric portion **410a** of each crankshaft **410** via a first roller bearing **418a**. As each crankshaft **410** rotates, the first eccentric portion **410a** eccentrically rotates. The eccentric rotation results in the first external gear **414** oscillatorily rotating while meshing with the internal tooth pins **417**.

(158) The first external gear **414** is sized such that it is slightly smaller than the inner diameter of the casing **422**. The first external gear **414** has first external teeth **414a**, a central through hole **414b**, a plurality of (for example, three) first eccentric portion insertion holes **414c**, and a plurality of (for example, three) shaft portion insertion holes **414d**. The first external teeth **414a** are shaped like smooth and continuous waves along the entire circumference of the external gear **414**.

(159) The central through hole **414b** is provided at the center of the first external gear **414** in the radial direction. The central through hole **414b** receives therein the input shaft **408** with a clearance therebetween.

(160) The three first eccentric portion insertion holes **414c** are arranged at regular intervals next to each other in the circumferential direction around the central through hole **414b** in the first external gear **414**. Each first eccentric portion insertion hole **414c** receives therein the first eccentric portion **410a** of each crankshaft **410** via the first roller bearing **418a**.

(161) The three shaft portion insertion holes **414d** are arranged at regular intervals next to each other in the circumferential direction around the central through hole **414b** in the first external gear **414**. The shaft portion insertion holes **414d** are positioned between the three first eccentric portion insertion holes **414c** in the circumferential direction. Each shaft portion insertion hole **414d** receives therein a corresponding one of the shaft portions **404c** with a clearance therebetween.

(162) The second external gear **416** is located in the closed space within the casing **422** and is attached to the second eccentric portion **410b** of each crankshaft **410** via a second roller bearing **418b**. The first and second external gears **414** and **416** are next to each other in the axial direction correspondingly to the first and second eccentric portions **410a** and **410b**. As each crankshaft **410** rotates, the second eccentric portion **410b** eccentrically rotates. The eccentric rotation results in the

second external gear **416** oscillatorily rotating while meshing with the internal tooth pins **417**.

(163) The second external gear **416** is sized such that it is slightly smaller than the inner diameter of the casing **422** and configured in the same manner as the first external gear **414**. In other words, the second external gear **416** has second external teeth **416a**, a central through hole **416b**, a plurality of (for example, three) second eccentric portion insertion holes **416c**, and a plurality of (for example, three) shaft portion insertion holes **416d**. These are configured in the same manner as the first external teeth **414a**, the central through hole **414b**, the first eccentric portion insertion holes **414c**, and the shaft portion insertion holes **414d**. Each second eccentric insertion hole **416c** receives therein the second eccentric portion **410b** of each crankshaft **410** via the second roller bearing **418b**.

(164) Each transmission gear **420** transmits the rotation of the input gear **408a** to the corresponding one of the crankshafts **410**. Each of the transmission gears **420** is fitted onto the mating portion **410c**, which is provided on one of the ends of the shaft body **412c** of the corresponding crankshaft **410**. Each transmission gear **420** is rotatable integrally with the crankshaft **410** around the same axis as the crankshaft **410**. Each of the transmission gears **420** has external teeth **420a** that mesh with the input gear **408a**.

(165) The following describes the materials used to make the constituents of the speed reducer **400** relating to the present embodiment shown in FIGS. **11** and **12**.

(166) In the speed reducer **400** relating to the present embodiment, the carrier **404** and the casing **422** are made of resin so that the speed reducer **400** can have a reduced weight. The inner sliding surface **448** serving as the outer ring of the main bearings **424** and **426** is made of a material having a higher thermal conductivity than the resin making the first and second carriers **404a** and **404b** forming the outer sliding surface **449** serving as the inner ring.

(167) Any material can be used to make the metal rings **444** and **445** serving as the outer ring of the main bearings **424** and **426** as long as the material exhibits a higher thermal conductivity than the resin of the carrier **404** serving as the inner ring and a higher strength than the resin of the carrier **404**. The material can be a metal material, a non-metal material or the like. The metal rings **444** and **445** relating to the present embodiment may be made of copper- or aluminum-based metals or alloys, or iron-based metals such as bearing steel and stainless steel. The internal tooth pins **417** may be made of the same material as the metal rings **444** and **445**. The input shaft **408**, the crankshafts **410**, the first roller bearings **418a**, the second roller bearings **418b**, the first crank bearings **412a**, the second crank bearings **412b**, the transmission gears **420** and the like may be formed from the same material as the metal rings **444** and **445**. Furthermore, the first and second external gears **414** and **416** may be made of the same material as the carrier **404**.

(168) In the speed reducer **400** relating to the present embodiment, the main bearings **424** and **426** have no heavy and metal constituents heavier than resins, except for the metal rings **444** and **445**. Thus, the speed reducer **400** can achieve further weight reduction.

(169) According to the above arrangement of the present embodiment, the heat transferred to the first and second metal rings **444** and **445** can be dissipated outside through the main bearings **424** and **426**, which are sliding bearings. Therefore, the speed reducer **400** can achieve improved heat dissipation. Due to the improved heat dissipation, the inner and outer sliding surfaces **448** and **449** can be saved from excessive temperature rise, which may melt the resin of the surfaces or cause the surfaces to stick to each other. In this manner, the speed reducer **400** can be prevented from failing. This can accordingly prevent the heat from being contained in the speed reducer **400**, thereby preventing problems to occur.

(170) On the inner periphery of the casing **422**, the first metal ring **444**, the internal tooth pins **417** and the second metal ring **445**, which are made of metals, are arranged adjacent to each other in the axial direction and in contact with each other. This arrangement can contribute to maintain the strength of the casing **422** along the entire length in the axial direction. In this manner, the speed reducer **400** can be prevented from failing and maintain sufficient strength. In addition, the above

arrangement can prevent a rise in the surface pressure of the tooth flanks of the external gears **414** and **416**, thereby reducing a loss in the lifetime of the external gears **414** and **416**.

(171) The present embodiment is not limited to the above, and can be modified and improved in various manners within a spirit of the invention. For example, while two oscillating external gears **414** and **416** are provided in the present embodiment, the present embodiment is not limited to such. For example, one or three or more external gears may be alternatively provided.

(172) In the present embodiment, the input shaft **408** is provided at the center of the carrier **404**, and the crankshafts **410** are arranged to surround the input shaft **408**. The present embodiment, however, is not limited to such. For example, it may also be possible to adopt a center crank type gear device in which the crankshafts **410** are arranged at the center of the carrier **404**. In this case, the input shaft **408** may be located at any position as long as the input shaft **408** can mesh with the transmission gears **420** mounted on the crankshafts **410**.

(173) According to the foregoing illustrative description, the casing **422** and the carrier **404** are made of resin, and the metal rings **444** and **445** are made of metals such as aluminum alloys in the present embodiment. As an alternative example, the casing **422** can be made of metals, and the carrier **404** can be made of metals. In this case, the metals can be aluminum alloys, but the present embodiment is not limited to such. The constituents of the speed reducer **400** can be made of any appropriately selected one of the materials that can achieve light weight and sufficient rigidity as long as the inner and outer sliding surfaces **448** and **449** are made of different materials such as resin and metals.

(174) Furthermore, the speed reducer **400** relating to the present embodiment can produce the same effects as the speed reducers relating to the above-described embodiments.

Eighth Embodiment

(175) The following describes a speed reducer relating to an eighth embodiment of the disclosure with reference to the accompanying drawings. FIG. **13** is a sectional view showing the speed reducer relating to the eighth embodiment along a direction extending along a main axis. In FIG. **13**, the reference numeral **500** indicates the speed reducer.

(176) The speed reducer **500** of the present embodiment is an eccentric oscillation type and includes an input shaft (an eccentric body) **512**, eccentric portions **503**, external gears **514** corresponding to the eccentric portions **503**, eccentric portion bearings **509**, a carrier **519**, and an internal gear **516** as shown in FIG. **13**.

(177) The input shaft **512** has an opening **512D** at the output-side end thereof in the axial direction extending along a central axis (the main axis) **5La**, as shown in FIG. **13**. The input shaft **512** has a protrusion **512A** at the input-side end thereof in the axial direction, with which a motor (not shown) is capable of engaging. The input shaft **512** is situated at the radial center of the speed reducer **500**. The input shaft **512** has a supporting portion **512B** formed at a position close to the output-side end and a supporting portion **512C** formed at a position close to the input-side end.

(178) The input shaft **512** is supported by the carrier **519** via a pair of bearings **534** and **536**. The bearing **534** is aligned with the supporting portion **512B** and supports the input shaft **512**. The bearing **536** is aligned with the supporting portion **512C** and supports the input shaft **512**. The rolling elements of the bearings **534** and **536** are balls. The rolling elements provide a gap (clearance), which is not shown, between the inner ring and the outer ring. The input shaft **512** has the eccentric portions **503** integrated therewith, which are sandwiched between the bearings **534** and **536**. With such arrangement, the input shaft **512** can serve as an eccentric shaft. In the direction extending along the central axis **5La**, the output side refers to the direction from the eccentric portions **503** to the bearing **534**. In the direction extending along the central axis **5La**, the input side refers to the direction from the eccentric portions **503** to the bearing **536**.

(179) The eccentric portions **503** include a first eccentric portion **503a**, a second eccentric portion **503b** and a third eccentric portion **503c**. The three or first, second and third eccentric portions **503a**, **503b** and **503c** are arranged next to each other in the direction extending along the central

axis **5La**. The first and third eccentric portions **503a** and **503c** (the two outer eccentric portions) are spaced apart along the central axis **5La**. Between the first and third eccentric portions **503a** and **503c**, the second eccentric portion (inner eccentric portion) **503b** is placed. In other words, in the direction extending along the central axis **5La**, the first eccentric portion **503a** and the third eccentric portion **503c** sandwich the second eccentric portion **503b** therebetween.

(180) The centers of the three eccentric portions **503a**, **503b** and **503c** are eccentric to the central axis **5La** of the input shaft **512** by the same amount. The three eccentric portions **503a**, **503b** and **503c** are arranged with a phase difference of 120 degrees in terms of eccentricity, which is obtained by dividing 360 degrees by three or the number of eccentric portions **503a**, **503b** and **503c**. The maximum eccentric positions of the eccentric portions **503a**, **503b** and **503c**, where their radial dimension relative to the central axis **5La** takes the largest value, are positioned differently from each other in the circumferential direction centered on the central axis **5La**.

(181) Specifically, in the clockwise (counter-clockwise) direction centered around the central axis **5La** of the input shaft **512**, relative to the maximum eccentric position of the first eccentric portion **503a**, the maximum eccentric position of the second eccentric portion **503b** is 120 degrees away. Furthermore, the maximum eccentric position of the third eccentric portion **503c** is 120 degrees away relative to the maximum eccentric position of the second eccentric portion **503b**. In addition, the maximum eccentric position of the first eccentric portion **503a** is 120 degrees away from the maximum eccentric position of the third eccentric portion **503c**.

(182) The eccentric portion bearings **509** are arranged around the outer periphery of the eccentric portions **503** and are configured to transmit the eccentric rotation of the eccentric portions **503**. The eccentric portion bearings **509** include a first eccentric portion bearing **509a**, a second eccentric portion bearing **509b** and a third eccentric portion bearing **509c**, which are corresponding to the eccentric portions **503a**, **503b** and **503c**. The three eccentric portion bearings **509a**, **509b** and **509c** each have a roller and a retainer that restricts the roller in the circumferential direction. None of the three eccentric portion bearings **509a**, **509b** and **509c** have inner or outer rings. Note that the term “roller” includes “needle roller” as used herein.

(183) The external gears **514** are attached to the outer periphery of the eccentric portions **503** via the eccentric portion bearings **509**. The eccentric portions **503** cause the external gears **514** to oscillatorily rotate. The external gears **514** include a first external gear **514a**, a second external gear **514b** and a third external gear **514c**, which are corresponding to the eccentric portions **503a**, **503b** and **503c**.

(184) The three external gears **514a**, **514b** and **514c** respectively include a plurality of inner pin holes **515a**, **515b** and **515c**. The inner pin holes **515a**, **515b** and **515c** penetrate the external gears **514a**, **514b** and **514c**, respectively. In the external gears **514a**, **514b** and **514c**, inner pins **540** with rotatable inner rollers **537** are fitted with a clearance.

(185) The carrier **519** includes a first carrier (shaft flange) **518** and a second carrier (hold flange) **520**. The first carrier **518** has the inner pins **540** integrated therewith. The first and second carriers **518** and **520** are coupled together using bolts **540a** to form a single part. The bolts **540a** are screwed in from the outside of the second carrier **520** and coupled to the inner pins **540**.

(186) The first carrier **518** is located on the output side with respect to the first external gear **514a**. The second carrier **520** is located on the input side with respect to the third external gear **514c**. The carrier **519** is supported on the casing **522** via the first and second main bearings **524** and **526**. The first carrier **518** is supported by the casing **522** via a first main bearing **524**. The second carrier **520** is supported by the casing **522** via a second main bearing **526**. In the following description, the first and second main bearings **524** and **526** may be simply referred to as the main bearings **524** and **526**.

(187) The main bearings **524** and **526** are both plain bearings constituted by an inner sliding surface **548** and an outer sliding surface **549**. The main bearings **524** and **526** respectively have first and second metal rings **544** and **545**. In the following description, the first and second metal rings **544**

and 545 may be simply referred to as the metal rings 544 and 545. The main bearing 524 has the first metal ring 544. The second main bearing 526 has the second metal ring 545. The first metal ring 544 is positioned on the output side with respect to the casing 522. The second metal ring 545 is positioned on the input side with respect to the casing 522. The first and second metal rings 544 and 545 are integrated into the casing 522.

(188) The outer periphery of the first metal ring 544 is fixedly attached to the inner periphery of the casing 522. The casing 522 has a depression on the inner periphery thereof for receiving therein the first metal ring 544. The first metal ring 544 is housed within the depression formed in the casing 522 in the axial direction. The first metal ring 544 serves as the outer ring of the first main bearing 524.

(189) The first metal ring 544, which serves as the outer ring of the first main bearing 524, is provided on the casing 522 on the output side. The inner peripheral surface of the first metal ring 544 constitutes the inner sliding surface 548 serving as the outer ring. The inner sliding surface 548 is in contact with the first carrier (shaft flange) 518. The first metal ring 544 is firmly fitted onto the casing 522, and the fit can be, for example, a clearance fit, an interference fit or a transition fit. The clearance may be determined by the difference in thermal expansion coefficient.

(190) The outer periphery of the second metal ring 545 is fixedly attached to the inner periphery of the casing 522. The casing 522 has a depression on the inner periphery thereof for receiving therein the second metal ring 545. The second metal ring 545 protrudes beyond the depression formed in the casing 522 toward the input side in the direction extending along the central axis 5La. The second metal ring 545 serves as the outer ring of the second main bearing 526.

(191) The second metal ring 545, which serves as the outer ring of the second main bearing 526, is provided on the casing 522 on the input side. The second metal ring 545 is exposed on the input side of the casing 522. The inner peripheral surface of the second metal ring 545 constitutes the inner sliding surface 548 serving as the outer ring. The inner sliding surface 548 is in contact with the second carrier (hold flange) 520. The second metal ring 545 is firmly fitted onto the casing 522, and the fit can be, for example, a clearance fit, an interference fit or a transition fit. The clearance may be determined by the difference in thermal expansion coefficient.

(192) The first main bearing 524 is arranged between the first carrier 518 and the casing 522. The second main bearing 526 is arranged between the second carrier 520 and the casing 522. The main bearings 524 and 526 have the outer sliding surface 549 serving as the inner ring. The outer sliding surface 549 is in contact with the inner sliding surface 548. The outer sliding surface 549 is formed on the outer peripheral surface of the carriers 518 and 520. The outer sliding surface 549 is movable while remaining slidable against the inner sliding surface 548 formed on the inner peripheral surface of the metal rings 544 and 545.

(193) As for the main bearings 524 and 526, the inner ring is the outer sliding surface 549, which is formed on the carriers 518 and 520, and the outer ring is the inner sliding surface 548, which is formed on the metal rings 544 and 545. The metal rings 544 and 545 are supported on the inner periphery of the casing 522.

(194) The inner and outer sliding surfaces 548 and 549 formed on the first and second metal rings 544 and 545 are positioned at substantially the same distance in the radial direction as are outer pins (internal tooth pins) 517. The first and second metal rings 544 and 545 are both in contact with the internal tooth pins 517 in the direction extending along the central axis 5La of the input shaft 512. Since the first and second metal rings 544 and 545 and the internal tooth pins 517 are in contact with each other in the direction extending along the central axis 5La, the speed reducer 500 can keep the strength along the entire length of the casing 522 in the direction extending along the central axis 5La.

(195) The inner sliding surface 548 serving as the outer ring of the main bearings 524 and 526 is made of a material having a higher thermal conductivity than the resin making the carriers 518 and 520 forming the outer sliding surface 549 serving as the inner ring.

(196) Any material can be used to make the metal rings **544** and **545** serving as the outer ring of the main bearings **524** and **526** as long as the material exhibits a higher thermal conductivity than the resin of the carrier **518** and **520** serving as the inner ring and a higher strength than the resin of the carriers **518** and **520**. The material can be a metal material, a non-metal material or the like. The metal rings **544** and **545** relating to the present embodiment may be made of copper- or aluminum-based metals, or iron-based metals such as bearing steel.

(197) The internal gear **516** includes the internal tooth pins **517** shaped like a circular cylinder and an internal gear body **516a** having pin grooves **516b** formed therein for rotatably supporting the internal tooth pins **517**. The internal gear body **516a** is integrated with the casing **522**. There is a small difference in the number of teeth between the internal gear **516** and the first external gear **514a**. There is a small difference in the number of teeth between the internal gear **516** and the second external gear **514b**. There is a small difference in the number of teeth between the internal gear **516** and the third external gear **514c**. An oil seal **533** is disposed between the casing **522** and the first carrier **518** and outside the metal ring **544**.

(198) As a motor, which is not shown, is driven, the input shaft **512** resultantly rotates. As a result, the eccentric portions **503** provided on the outer periphery of the input shaft **512** eccentrically rotate integrally with the input shaft **512**. The rotation of the eccentric portions **503** causes the external gears **514a**, **514b** and **514c**, which correspond to the eccentric portions **503a**, **503b** and **503c**, to attempt to oscillatorily rotate around the input shaft **512**. Here, the external gears **514a**, **514b** and **514c** are restricted by the internal gear **516** from rotating on their own axis and are thus only allowed to oscillate while keeping in contact with the internal gear **516**.

(199) The inner pin holes **515a**, **515b** and **515c** and the inner pins **540** (and the inner rollers **537**) absorb the oscillation component. As a result, the first, second and third external gears **514a**, **514b** and **514c** rotate relative to the internal gear **516**, which is stationary, by the amount determined by the difference in the number of teeth between the external gears **514a**, **514b** and **514c** and the internal gear **516**. In other words, it is only the component of the rotation on their own axis resulting from the difference in the number of teeth between the first to third external gears **514a** to **514c** and the internal gear **516** that is transmitted to the carrier **519**.

(200) The following describes the materials used to make the constituents of the speed reducer **500** relating to the present embodiment shown in FIG. **13**.

(201) In the speed reducer **500** relating to the present embodiment, the carrier **519**, the casing **522**, the first external gear **514a**, the second external gear **514b**, and the third external gear **514c** are made of resin so that the speed reducer **500** can have a reduced weight. The inner sliding surface **548** serving as the outer ring of the main bearings **524** and **526** may be made of a material having a higher thermal conductivity than the resin making the first and second carriers **518** and **520** forming the outer sliding surface **549** serving as the inner ring.

(202) Any material can be used to make the metal rings **544** and **545** serving as the outer ring as long as the material exhibits a higher thermal conductivity than the resin of the carrier **519** serving as the inner ring and a higher strength than the resin of the carrier **519**. The material can be a metal material, a non-metal material or the like. The metal rings **544** and **545** relating to the present embodiment may be made of copper- or aluminum-based metals or alloys, or iron-based metals such as bearing steel and stainless steel. The internal tooth pins **517** may be made of the same material as the metal rings **544** and **545**. The input shaft **512**, the eccentric portion bearings **509** and the bearings **534** and **536** may be made of the same material as the metal rings **544** and **545**.

(203) In the speed reducer **500** relating to the present embodiment, the main bearings **524** and **526** have no heavy and metal constituents heavier than resin, except for the metal rings **544** and **545**. Thus, the speed reducer **500** can achieve further weight reduction.

(204) According to the above arrangement of the present embodiment, the heat transferred to the first and second metal rings **544** and **545** can be dissipated outside through the main bearings **524** and **526**, which are sliding bearings. Accordingly, the speed reducer **500** can accomplish improved

heat dissipation. Due to the improved heat dissipation, the inner and outer sliding surfaces **548** and **549** can be saved from excessive temperature rise, which may melt the resin of the surfaces or cause the surfaces to stick to each other. In this manner, the speed reducer **500** can be prevented from failing. This can accordingly prevent the heat from being contained in the speed reducer **500**, thereby preventing problems to occur.

(205) On the inner periphery of the casing **522**, the first metal ring **544**, the internal tooth pins **517** and the second metal ring **545**, which are made of metals, are arranged adjacent to each other in the axial direction and in contact with each other. This arrangement can contribute to maintain the strength of the casing **522** along the entire length in the axial direction. This can prevent the failure of the speed reducer **500** and maintain sufficient strength.

(206) In the present embodiment shown in FIG. **13**, the first carrier **518** serves as an output shaft to a destination machine, which is not shown. The carrier **519** includes the second carrier **520**, so that the inner pins **540** are held on both sides, but the carrier **519** may be configured to hold the inner pins on one side. Furthermore, three or more eccentric portions are arranged with a phase difference of 360 degrees/(number of eccentric portions) in terms of eccentricity, but the present embodiment is not limited to such. Furthermore, the present embodiment is not limited to such a configuration that the length in the axial direction of the two outer ones of the three or more eccentric portions that are positioned outermost in the axial direction is less than the length in the axial direction of the inner eccentric portion that is positioned inside the outer eccentric portions.

(207) The speed reducer **500** relating to the present embodiment can produce the same effects as the speed reducers relating to the above-described embodiments.

Ninth Embodiment

(208) The following describes a speed reducer relating to a ninth embodiment of the disclosure with reference to the accompanying drawings. FIG. **14** is a sectional view showing the speed reducer relating to the ninth embodiment along a direction extending along a main axis. FIG. **15** is a sectional view along a line XV-XV in FIG. **14**. In FIGS. **14** and **15**, the reference numeral **600** indicates the speed reducer.

(209) The speed reducer **600** relating to the ninth embodiment is an eccentric oscillation speed reducer having an external gear and an internal gear meshing with each other. Oscillating the external gear causes one of the internal and external gears to rotate on its own axis, so that the generated rotational component is output to a driven device through an output member.

(210) The speed reducer **600** relating to the ninth embodiment includes, as shown in FIGS. **14** and **15**, an input shaft **612**, external gears **614**, an internal gear **616**, a first carrier **618**, a second carrier **620**, a casing **622**, a first main bearing **624**, a second main bearing **626** and carrier pins **638**.

Hereinafter, the expression “axial direction” refers to the direction extending along the central axis (the main axis) **6La** of the internal gear **616**, and the expressions “circumferential direction” and “radial direction” respectively refer to the circumferential and radial directions of the circle centered on the central axis **6La**. In the following description, for convenience, the input side refers to one side in the axial direction (the right side in FIG. **14**) and the non-input side or output side refers to the other side (the left side in FIG. **14**). In the following description, the first and second carriers **618** and **620** may be simply referred to as the carriers **618** and **620**. Furthermore, the first and second main bearings **624** and **626** may be simply referred to as the main bearings **624** and **626**.

(211) The input shaft **612** is rotatable around the center line when acted upon by rotational power input from a drive source. The speed reducer **600** relating to the ninth embodiment is, as shown in FIGS. **14** and **15**, a center crank type, where the rotational center line of the input shaft **612** is aligned with the center line **6La** of the internal gear **616**. The drive source is, for example, a motor, a gear motor or an engine.

(212) The input shaft **612** is an eccentric shaft having a plurality of eccentric portions **612a** for oscillating the external gears **614**. Having such a configuration, the input shaft (eccentric body) **612**

is sometimes referred to as a crankshaft. The axial center of the eccentric portions **612a** is eccentric to the rotational center line of the input shaft **612**. In the present embodiment, two eccentric portions **612a** are adjacent to each other on the input shaft **612**. The adjacent eccentric portions **612a** are 180° out of phase in terms of eccentricity.

(213) The input shaft **612** is supported on the input side by the second carrier **620** via an input shaft bearing **634** and on the non-input side by the first carrier **618** via an input shaft bearing **634**. The input shaft **612** is rotatably supported by the first and second carriers **618** and **620**. The input shaft bearings **634** are not limited to any particular configurations, but can be, for example, ball bearings with spherical rollers.

(214) The internal gear **616** meshes with the external gears **614**. According to the present embodiment, the internal gear **616** includes an internal gear body **616a** integrated into the casing **622**, and outer pins (internal tooth pins) **617** disposed in pin grooves **616b** formed in the internal gear body **616a** at intervals in the circumferential direction. The outer pins **617** are cylindrical pin members rotatably supported on the internal gear body **616a**. The outer pins **617** have the same radial dimension along the entire length in the axial direction. The outer pins **617** form the internal teeth of the internal gear **616**. The number of outer pins **617** (the number of internal teeth) of the internal gear **616** is slightly (in this example, by one) greater than the number of external teeth of the external gear **614**.

(215) The internal gear body **616a** and the casing **622** are made of resin. Although the internal gear body **616a** and the casing **622** can be made of a variety of types of resin, the internal gear body **616a** and the casing **622** are made of polyacetal (POM) in this example. The present embodiment, however, is not limited to such, and the internal gear body **616a** and the casing **622** may be made of other resin than POM, for example, polyetheretherketone (PEEK).

(216) The resin used to make the internal gear body **616a**, the casing **622** and other constituent parts of the present embodiment may be a resin containing reinforcing fibers such as glass or carbon fibers, a resin without reinforcing fibers, a lamination of a base material such as paper or fabric impregnated with a resin. The resin used for the constituent parts of the present embodiment may be a resin having thermally conductive fillers mixed therein.

(217) In the speed reducer **600**, the outer pins **617** may be made of a material that has a higher thermal conductivity [W/(m.Math.K)] than the resin material used for the internal gear body **616a**.

(218) The outer pins **617** can be made of any material as long as it exhibits higher thermal conductivity and rigidity than the resin used to make the internal gear body **616a** and can be made of metal, highly thermally conductive resin, non-metal materials and the like. The outer pins **617** may be, for example, made of a resin having carbon nanotubes (CNTs), boron nitride nanotubes (BNNTs) mixed therein. The outer pins **617** relating to the present embodiment, shown in FIGS. **14** and **15**, may be made of iron-based metals such as bearing steel.

(219) The outer pins **617** can be solid or hollow. The outer pins **617** may have a multi-layered structure with a core material being wrapped in a surface material. For example, one of the core and surface materials of the multi-layered structure of the outer pins **617** may be made of iron-based metals, and the other may be made of copper- or aluminum-based metals. Such a structure can eliminate the trade-off between the mechanical and thermal characteristics. As yet another example of the core and surface materials of the multi-layered structure of the outer pins **617**, one of them may be made of metals and the other may be made of resin. The outer pins **617** may be made of sintered metals and ceramics.

(220) The external gears **614** are provided in a one-to-one correspondence with the eccentric portions **612a**. The external gears **614** are rotatably supported on the eccentric portions **612a** via eccentric bearings **630**. As shown in FIG. **15**, the external gears **614** have a plurality of carrier pin holes (inner pin holes) **639** provided off the central axis of the external gears **614**, arranged at equal intervals, and next to each other in the circumferential direction. The carrier pin holes **639** receive therein carrier pins (inner pins) **638**. The carrier pin holes **639** all have the same diameter. The

radial dimension of the carrier pin holes **639** is greater than the radial dimension of the carrier pins **638**.

(221) The external gears **614** are made of resin, like the internal gear body **616a**. The external gears **614** can be made of a variety of resin. The external gears **614** are located closer to the input shaft **612** than is the internal gear body **616a**. The external gears **614** may be made of, for example, PEEK. The present embodiment, however, is not limited to such, and the external gears **614** may be made of other resin than PEEK, for example, POM.

(222) The carrier pin holes **639** are at the same distance from the center in the radial direction and shaped like a circle. Corrugated teeth are formed on the outer periphery of the external gears **614**. The external gears **614** oscillate within a plane to which the central axis **6La** is normal, such that the corrugated teeth move while keeping touching the internal gear **616**. A gap is provided between the carrier pins **638** and the carrier pin holes **639** to absorb the oscillating component of the external gears **614**. The carrier pins **638** partially touch the inner wall surface of the carrier pin holes **639**.

(223) The carriers **618** and **620** sandwich the external gears **614** therebetween in the axial direction. The first carrier (shaft flange) **618** is located on the non-input-side lateral portion of the external gears **614**. The second carrier (hold flange) **620** is located on the input-side lateral portion of the external gears **614**. The first and second carriers **618** and **620** are rotatably supported by the casing **622** via the first and second main bearings **624** and **626**. The first carrier (shaft flange) **618** is rotatably supported by the casing **622** via the first main bearing **624**. The second carrier (hold flange) **620** is rotatably supported by the casing **622** via the second main bearing **626**.

(224) The carriers **618** and **620** are shaped like a disc as a whole. The first carrier **618** rotatably supports the input shaft **612** via the input shaft bearing **634**. The second carrier **620** rotatably supports the input shaft **612** via the input shaft bearing **634**. The carrier pins (inner pins) **638** are coupled to the first carrier (shaft flange) **618** and to the second carrier (hold flange) **620** with bolts **638a**. The bolts **638a** are made of rigid materials such as iron-based metals.

(225) The first and second carriers **618** and **620** are coupled to each other via the carrier pins **638**. The carrier pins **638** are positioned off the axial center of the external gears **614** in the radial direction and penetrate the external gears **614** in the axial direction. In the present embodiment, the carrier pins **638** are separate parts from the carriers **618** and **620**. The present embodiment, however, is not limited to such. As another example, some of the carrier pins **638** may be integrated with the carriers **618** and **620**.

(226) One of the first carrier **618** and the casing **622** serves as an output member for outputting rotational power to a driven device, and the other serves as a stationary member fixedly attached to an external member for supporting the speed reducer **600**. The output member is rotatably supported on the stationary member via the main bearings **624** and **626**. The non-input-side end surface of the first carrier **618** may have a driven member, which is to be rotated by the speed reducer **600**, coupled thereto with bolts or the like. Alternatively, the outer peripheral flange of the casing **622** may have a driven member, which is to be rotated by the speed reducer **600**, coupled thereto with bolts or the like.

(227) The casing **622** is shaped like a hollow tube as a whole. The internal gear **616** is provided on the inner periphery of the casing **622**. A flange or the like may be provided on the outer periphery of the casing **622**. The casing **622** has pin grooves **616b** supporting the outer pins **617**, which extend along the central axis **6La**. The outer pins **617** extend beyond the pin grooves **616b** in the direction extending along the central axis **6La**. In other words, the length of the outer pins **617** in the direction extending along the central axis **6La** is greater than the length of the pin grooves **616b** in the direction extending along the central axis **6La**. The outer pins **617** protrude outward beyond the pin grooves **616b** in the direction extending along the central axis **6La**.

(228) The carriers **618** and **620** have circumferential recesses **646** and **647** housing therein the protruding portions of the outer pins **617** beyond the pin grooves **616b**. The circumferential

recesses **646** and **647** are continuous in the circumferential direction of the carriers **618** and **620**. The circumferential recesses **646** and **647** have the same sectional shape along the entire length in the circumferential direction. The circumferential recesses **646** and **647** have a rectangular cross-section when cut along the radial direction. The sectional shape of the circumferential recesses **646** and **647** in the radial direction is compatible with the sectional shape of the outer pins **617** in the axial direction. The circumferential recesses **646** and **647** each have a flat annular surface **649a** extending along the radial direction and a cylindrical outer peripheral surface **649b** connected to the annular surface **649a**.

(229) As the carriers **618** and **620** rotate relative to the casing **622** around the central axis **6La**, the outer pins **617** protruding beyond the pin grooves **616b** are allowed to move in the circumferential direction within the circumferential recesses **646** and **647**. The circumferential recesses **646** and **647** formed in the carriers **618** and **620** serve as the inner ring of the main bearings **624** and **626**.

(230) The main bearings **624** and **626** have the outer pins **617** and the pin grooves **616b** serving as retainers for defining the positions of the outer pins **617** in the circumferential direction. The internal gear body **616a** serves as the outer ring of the main bearings **624** and **626**. This means that the main bearings **624** and **626** of the present embodiment are plain bearings formed by the outer pins **617**. The first main bearing **624** is arranged between the first carrier **618** and the casing **622**. The second main bearing **626** is arranged between the second carrier **620** and the casing **622**.

(231) The first main bearing **624** is constituted by the pin grooves **616b** formed in the casing **622**, the outer pins **617** and the circumferential recess **646** provided in the outer periphery of the first carrier **618** on the input side. In the circumferential recess **646**, the end surface **617b** of the outer pins **617** is slightly separated from or in contact with the annular surface **649a**. In the circumferential recess **646**, the peripheral surface **617c** of the outer pins **617** is in contact with the outer peripheral surface **649b**.

(232) As the first carrier **618** rotates relative to the casing **622** around the central axis **6La**, the output-side end surface **617b** of the outer pins **617** moves without contacting the annular surface **649a** or while remaining in sliding contact with the annular surface **649a**. The peripheral surface **617c** of the outer pins **617** near the end surface **617b** moves while remaining rolling contact with the outer peripheral surface **649b** as the first carrier **618** rotates relative to the casing **622** around the central axis **6La**. The peripheral surface **617c** of the outer pins **617** remains in contact with the outer peripheral surface **649b** along a line extending along the central axis **6La**.

(233) The second main bearing **626** is constituted by the pin grooves **616b** formed in the casing **622**, the outer pins **617**, and the circumferential recess **647** provided in the outer periphery of the second carrier **620** on the output side. In the circumferential recess **647**, the end surface **617b** of the outer pins **617** is slightly separated from or in contact with the annular surface **649a**. In the circumferential recess **647**, the peripheral surface **617c** of the outer pins **617** is in contact with the outer peripheral surface **649b**.

(234) As the second carrier **620** rotates relative to the casing **622** around the central axis **6La**, the input-side end surface **617b** of the outer pins **617** moves without contacting the annular surface **649a** or while remaining in sliding contact with the annular surface **649a**. The peripheral surface **617c** of the outer pins **617** near the end surface **617b** moves while remaining rolling contact with the outer peripheral surface **649b** as the second carrier **620** rotates relative to the casing **622** around the central axis **6La**. The peripheral surface **617c** of the outer pins **617** remains in contact with the outer peripheral surface **649b** along a line extending along the central axis **6La**.

(235) The length of the portion of the peripheral surface **617c** that is in contact with the outer peripheral surface **649b** in the direction extending along the central axis **6La** is the same among the outer pins **617**. The sites where the peripheral surfaces **617c** of the outer pins **617** are in contact with the outer peripheral surface **649b** are at equal intervals in the circumferential direction. The outer pins **617** are separated away from each other in the circumferential direction by a distance, which is reliably defined by the pin grooves **616b** separated away from each other in the

circumferential direction by a distance.

(236) In the main bearings **624** and **626**, the respective ends of each outer pin **617** in the direction extending along the central axis **6La** receive a radially outward force from the circumferential recesses **646** and **647**. In addition, the middle portion of each outer pin **617** in the direction extending along the central axis **6La** receives a radially inward force from the pin groove **616b**. This means that the main bearings **624** and **626** support the casing **622** and the carriers **618** and **620** such that the casing **622** can rotate relative to the carriers **618** and **620** along the entire circumference.

(237) The distance between the annular surface **649a** of the circumferential recess **646** and the annular surface **649a** of the circumferential recess **647** in the direction extending along the central axis **6La** is slightly greater than or substantially equal to the axial dimension of the outer pins **617**. Furthermore, the outer peripheral surface **649b** is at a given distance in the radial direction from the central axis **6La** and the given distance is equal to the value obtained by subtracting the radial dimension of the outer pins **617** from the maximum radial dimension of the pin grooves **616b**.

(238) The radial dimension of the annular surface **649a** can be equal to the radial dimension of the outer pins **617**. In this case, the outer peripheral surface of the carriers **618** and **620** has the same radial dimension as the cylindrical surface obtained by connecting together the radially outermost positions of the outer pins **617**. Here, all of the end surfaces **617b** of the outer pins **617** can touch the annular surface **649a**. The radial dimension of the annular surface **649a** can be smaller than the radial dimension of the outer pins **617**. In this case, the cylindrical surface obtained by connecting together the radially outermost radial positions of the outer pins **617** has a greater radial dimension than the outer peripheral surface of the carriers **618** and **620**. Furthermore, the length of the outer peripheral surface **649b** in the axial direction is substantially equal to the length, in the axial direction, of the peripheral surface **617c** of the outer pins **617** in contact with the outer peripheral surface **649b**.

(239) In the main bearings **624** and **626**, the outer pins **617** are made of a thermally conductive material having a higher resistance against wear than the resin making the outer ring, which is constituted by the pin grooves **616b** formed in (the inner peripheral surface **648** of) the casing **622**, and the inner ring, which is constituted by the outer peripheral surface **649b** formed in the carriers **618** and **620**. Specifically, any material can be used to make the outer pins **617** as long as the material exhibits a higher thermal conductivity than the resin of the casing **622** and the carriers **618** and **620** and a higher strength than the resin of the carriers **618** and **620**. The material can be a metal material, a non-metal material or the like. The outer pins **617** relating to the present embodiment shown in FIGS. **14** and **15** may be made of copper- or aluminum-based metals or alloys, or iron-based metals such as bearing steel.

(240) In the main bearings **624** and **626**, the outer pins **617** are sandwiched between (i) the pin grooves **616b** corresponding to the inner peripheral surface **648** serving as the outer ring and (ii) the outer peripheral surface **649b** of the circumferential recesses **646** and **647** serving as the inner ring. In other words, the outer pins **617** are sandwiched between (i) the casing **622** and (ii) the carriers **618** and **620**.

(241) The outer pins **617** protruding beyond the pin grooves **616b** in the direction extending along the central axis **6La** also constitute the internal gear **616**. Therefore, the main bearing **624**, the internal gear **616** and the main bearing **626** are arranged adjacent to each other along the central axis **6La**. This arrangement can contribute to maintain the strength of the speed reducer **600** along the entire length in the axial direction.

(242) The carrier pins (inner pins) **638** are inserted into the carrier pin holes (inner pin holes) **639**, which are formed in the external gears **614**, with a gap therebetween. The carrier pins **638** are fitted at one end thereof into recesses **618c** in the first carrier **618** and at the other end thereof into recesses **620c** in the second carrier **620**. The carrier pins **638** are fixedly inserted into the recesses **618c** and **620c** with bolts **638a**. The carrier pins **638** may be press-fitted in the recesses **618c** and

620c, in which case the carrier pins **638** are not secured using bolts or the like.

(243) The carrier pins (inner pins) **638** are in contact with part of the carrier pin holes **639**, which are formed in the external gears **614**. With such arrangement, the carrier pins **638** restrict the external gears **614** from rotating on their own axis and only allows the external gears **614** to oscillate. The carrier pins **638** serve as a coupling member playing a role in transmitting power between the first and second carriers **618** and **620** and the external gears **614**. Some of the carrier pins **638** may not be in contact with the carrier pin holes **639** formed in the external gears **614**. In this case, the carrier pins **638** do not play a role in restricting the external gears **614** from rotating on their own axis. Accordingly, the carrier pins **638** serve as a coupling member only contributing to couple together the first and second carriers **618** and **620** in this case.

(244) Speed reducers have seen an increase in its applications including collaborative robots working closely with people. In order to expand the applications, it is desired to reduce the weight and noise of the speed reducers. Conventional speed reducers are constituted by parts made of iron-based metals. In order to reduce the weight, the constituents may be made of materials with low specific gravity. Suitable materials with low specific gravity include resin. On the other hand, if the components are made of resin, this may compromise the strength and rigidity. Accordingly, the speed reducers may possibly experience problems or break down. If the components are made of resin, the components may experience temperature rise due to poorer heat dissipation and resultantly shortened lifetime. For this reason, the materials of the components are desirably selected in consideration of the purposes of unchanged strength, weight reduction and heat dissipation. In particular, the weight reduction may lead to compromised strength, which needs to be avoided.

(245) In the speed reducer **600**, many of its components can be made of resin to reduce weight. The casing **622**, the carriers **618** and **620** and the external gears **614** account for a large portion of the volume of the components of the speed reducer **600**. If these components are made of resin, the speed reducer **600** can achieve significant weight reduction. In consideration of the strength and heat dissipation, the input shaft **612**, the carrier pins (inner pins) **638**, the bolts **638a**, the eccentric bearings **630**, the input shaft bearings **634**, and the outer pins **617** are preferably made of metals.

(246) Furthermore, the main bearings **624** and **626** of the speed reducer **600** are constituted by the outer pins **617**, instead of having separate bearings, to rotatably support the casing **622** and the carriers **618** and **620**. This allows the speed reducer **600** to achieve further weight reduction. The main bearings **624** and **626** have no metal constituents except for the outer pins **617**. Thus, the speed reducer **600** can achieve further weight reduction.

(247) In the main bearings **624** and **626**, the peripheral surface **617c** of the outer pins **617** moves while remaining rolling contact with the outer peripheral surface **649b** as the carriers **618** and **620** rotate relative to the casing **622** around the central axis **6La**. This can reduce the friction and improve the reliability of the operation when compared with the case where the surfaces are in sliding contact. Furthermore, the annular surface **649a** and the outer peripheral surface **649b** are made of resin, and the outer pins **617** are made of metals. In this way, the heat generated from these components touching each other can be efficiently released to improve heat dissipation.

(248) In the speed reducer **600**, a large amount of heat is often generated around the main bearings **624** and **626**. In addition, a large amount of heat is often generated around the input shaft **612**, which is rotatable at a relatively high speed. Furthermore, the speed reducer **600** may malfunction unless the carrier pins (inner pins) **638** and the external gears **614** maintain sufficient strength. As noted, the speed reducers experience a large temperature increase if the heat generated inside is dissipated outside less efficiently. The resin components experience a sudden drop in rigidity and strength as the temperature rises. Accordingly, the continued use of the resin components may probably result in breakage.

(249) To deal with this, one of the members that are movable relative to each other is made of resin, the other member is desirably made of a material exhibiting a higher resistance against wear and a

higher thermal conductivity [W/(m.Math.K)] than the resin member. In this way, the heat generated inside can be dissipated outside in a better manner than when the other member is made of a low thermal conductivity material. Furthermore, the members can achieve a longer lifetime than when the other member is made of a low wear-resistance material. Furthermore, the noise can be reduced.

(250) The material used to make the outer pins **617** constituting the main bearings **624** and **626** may be made of any material as long as it exhibits higher wear-resistance and higher thermal conductivity than the resin used to make the outer peripheral surface **649b** and the annular surface **649a** serving as the inner ring. The material can be metal, non-metal, highly rigid, or highly thermally conductive materials. The outer pins **617**, which constitute the main bearings **624** and **626** of the present embodiment, may be made of iron- and aluminum-based metals such as bearing steel, light metals such as aluminum, magnesium, beryllium, and titanium, or composites thereof. Alternatively, the outer pins **617** may be made of ceramics. As the other components such as the carriers **618** and **620** and the casing **622** are made of resin, the speed reducer **600** can achieve both light weight and sufficient mechanical strength.

(251) The input shaft **612** and the eccentric bearings **630** and the input shaft bearings **634**, which are located in contact with the input shaft **612**, receive high-speed rotation input thereto before it is slowed down. This means that the input shaft **612**, the eccentric bearings **630** and the input shaft bearings **634** experience a relatively large temperature rise, and they allow a smaller number of rotations input thereto if they exhibit low resistance against heat. For this reason, the input shaft **612**, the input shaft bearings **634** and the eccentric bearings **630** may be made of metals, such as iron-based metals. This can prevent a drop in the allowable number of rotations input into them. Since the input shaft **612** is subjected to large torsional stress, the input shaft **612** is preferably made of a more rigid material than the carriers **618** and **620**. The input shaft **612** may be made of iron-based metals with a higher torsional strength than aluminum. The iron-based metals can include carbon steel, bearing steel, and stainless steel, any of which can be used depending on the desired characteristics.

(252) The carrier pins (inner pins) **638** are desirably highly rigid in order to reliably perform the transfer of the rotation to the external gears **614**. The carrier pins **638** are desirably highly rigid in order to reliably achieve sufficient coupling strength between the first and second carriers **618** and **620**. Considering this, the carrier pins **638** and the bolts **638a** may be made of metals. In this example, the carrier pins **638** may be made of iron-based metals.

(253) The following now describes how the speed reducer **600** configured as described above operates. As rotational power is transmitted from the driving unit to the input shaft **612**, the eccentric portions **612a** of the input shaft **612** rotate around the central axis **6La** of the input shaft **612**. This causes the eccentric portions **612a** to oscillate the external gears **614**. The external gears **614** oscillate in such a manner that their own axial center rotates around the rotational center line of the input shaft **612**. As the external gears **614** oscillate, the meshing between the external gears **614** and the outer pins **617** of the internal gear **616** takes place at sequentially shifted positions. Accordingly, every time the input shaft **612** rotates once, either the external gears **614** or the internal gear **616** rotates on its own axis by an amount determined by the difference between the number of teeth of the external gears **614** and the number of outer pins **617** of the internal gear **616**. In the present embodiment shown in FIGS. **14** and **15**, the rotation of the external gears **614** on their own axis results in reduced rotations being output through the first carrier **618** or casing **622**.

(254) Here, in the main bearings **624** and **626**, the outer pins **617** protruding beyond the pin grooves **616b** move while remaining rolling relative to the circumferential recesses **646** and **647** as the carriers **618** and **620** rotate relative to the casing **622** around the central axis **6La**. The peripheral surface **617c** of the outer pins **617** is in contact with the outer peripheral surface **649b** along a line extending along the central axis **6La**.

(255) In the speed reducer **600** relating to the present embodiment, the outer pins **617** serve both as

the internal tooth pins or the internal teeth of the internal gear **616** and as the rollers of the main bearings **624** and **626**. This allows the speed reducer **600** to be made up by a smaller number of parts, thereby achieving size and weight reduction. In particular, since no separate members need to be provided outside the outer pins **617** as part of the main bearings, the speed reducer **600** can have a reduced thickness. In addition, since the outer pins **617** are arranged near the outer periphery of the speed reducer **600** such that they cover a large part of the speed reducer **600** in the thickness direction, the speed reducer **600** can be strong enough to prevent deformation, which results in malfunctions.

(256) The speed reducer **600** relating to the present embodiment has no heavy and metal constituents heavier than resins, except for the outer pins **617**. Thus, the speed reducer **600** can achieve further weight reduction. Since one of the members is made of resin and the other is made of a material exhibiting higher wear-resistance and higher thermal conductivity than the resin, the members can achieve reduced weight, increased rigidity, enhanced heat dissipation and improved reliability of operation.

(257) The speed reducer **600** relating to the present embodiment can produce the same effects as the speed reducers relating to the above-described embodiments.

Tenth Embodiment

(258) The following describes a speed reducer relating to a tenth embodiment of the disclosure with reference to the accompanying drawings. FIG. **16** is a sectional view showing the speed reducer relating to the tenth embodiment along the axial direction. The present embodiment is different from the ninth embodiment in terms of the outer pins. The other components relating to the tenth embodiment have counterparts in the ninth embodiment, and they are assigned with the same reference signs and not described now.

(259) In the speed reducer **600** relating to the present embodiment, as shown in FIG. **16**, the outer pins **617** in the main bearings **624** and **626** each have diameter-enlarged portions **617f** at the respective ends. The diameter-enlarged portions **617f** are formed in the portion of each outer pin **617** that protrude beyond the pin groove **616b**. The portion of each outer pin **617** that is housed in the pin groove **616b** is configured in the same manner as in the ninth embodiment. The radial dimension of the diameter-enlarged portions **617f** is greater than the radial dimension of the portion of the outer pin **617** that is housed in the pin groove **616b**. The radial dimension of the diameter-enlarged portions **617f** remains unchanged along the entire length of the diameter-enlarged portions **617f** in the direction extending along the central axis **6La**. The diameter-enlarged portions **617f** at the respective ends of each outer pin **617** are both shaped in the same manner.

(260) The end surfaces **617b** of the diameter-enlarged portions **617f** of each outer pin **617** are slightly separated from or in contact with the annular surfaces **649a**. The peripheral surfaces **617c** of the diameter-enlarged portions **617f** touch the outer peripheral surfaces **649b**. As a result, the circumferential recesses **646** and **647** are positioned closer to the central axis **6La** in the radial direction. For similar reasons, the annular surfaces **649a** have a greater radial dimension.

(261) The circumferential recess **646** has a greater size in the radial direction than in the ninth embodiment, due to the sectional shape of the diameter-enlarged portion **617f**. The peripheral surface **617c** of the diameter enlarged portion **617f** moves while remaining rolling contact with the outer peripheral surface **649b** as the carriers **618** and **620** rotate relative to the casing **622** around the central axis **6La**. The peripheral surface **617c** of the outer pins **617** remains in contact with the outer peripheral surface **649b** along a line extending along the central axis **6La**.

(262) In the speed reducer **600** relating to the present embodiment shown in FIG. **16**, the peripheral surface **617c** of the diameter-enlarged portion **617f** has a greater radial dimension than the peripheral surface **617c** of the ninth embodiment. Here, if the outer pins **617** rotate at the same rate, the circumferential speed of the peripheral surface **617c** increases proportionally to the radial dimension. It is presupposed that the outer pins **617** rotate at a constant speed when the outer pins **617** serve as the internal tooth pins of the internal gear **616** and come into contact with the external

gears **614** as a result of the rotation of the carriers **618** and **620** relative to the casing **622** around the central axis **6La**.

(263) While rotating, the peripheral surface **617c** may touch the outer peripheral surface **649b** of the circumferential recesses **646** and **647**. As a result, if the outer pins **617** rotate in an ideal manner, the peripheral surface **617c** establishes a rolling contact with the outer peripheral surface **649b**. However, if the circumferential speed of the outer pins **617** is too different from the circumferential speed of the outer peripheral surface **649b**, the peripheral surface **617c** may unfortunately establish a sliding contact with the outer peripheral surface **649b**. The sliding contact may increase friction, generate more heat, shorten the lifetime, increase the noise, result in a higher failure rate, when compared with the rolling contact. Therefore, there is a demand to prevent these.

(264) In the speed reducer **600** relating to the present embodiment shown in FIG. **16**, the outer pins **617** are thicker as they have the diameter-enlarged portions **617f**. This can prevent the sliding contact from being established between the peripheral surface **617c** and the outer peripheral surface **649b** and help maintaining the rolling contact. In this manner, the circumferential speed of the outer pins **617** can be controlled. In other words, since the outer pins **617** have the diameter-enlarged portions **617f**, they may be less affected by the sliding resistance produced by the carriers **618** and **620**. In this manner, the outer pins **617** can be rotated mainly by the contact resistance produced between them and the external gears **614**.

(265) This can align the large number of outer pins **617** in terms of the number of rotations, so that the carriers **618** and **620** can rotate relative to the casing **622** around the central axis **6La** more reliably. Note that the radial dimension of the carriers **618** and **620** and casing **622** and their relative rotation around the central axis **6La** may determine the favorable number of rotations of the outer pins **617**.

(266) As described above, since the outer pins **617** have the diameter-enlarged portions **617f**, the circumferential speed of the peripheral surface **617c**, which is in contact with the outer peripheral surface **649b**, increases. This means that, even if the outer pins **617** rotate at the same speed, the circumferential speed of the outer pins **617** can be increased toward the circumferential speed of the outer peripheral surface **649b**. In this manner, the circumferential speed of the outer pins **617** can be easily aligned with the speed of the relative rotation between the carriers **618** and **620** and the casing **622**. Accordingly, the present embodiment can adjust the number of rotations of the large number of outer pins **617** more efficiently.

(267) Stated differently, in the speed reducer **600** relating to the present embodiment, the circumferential speed is higher on the peripheral surface **617c** of the diameter-enlarged portions **617f** than at a site in contact with the pin grooves **616b**. Since the outer pins **617** are in contact with the outer peripheral surface **649b** while rotating at a high speed, it is preferable that the circumferential speed of the peripheral surface **617c** abutting the outer peripheral surface **649b** is ideally the same among all of the outer pins **617**.

(268) Here, the variation in the speed among the outer pins **617**, in other words, the variation in the circumferential speed can be controlled or reduced more effectively if the relative speed of the outer pins **617**, which are in contact with the outer peripheral surface **649b**, is high. For this reason, the diameter-enlarged portions **617f** can help align the large number of outer pins **617** in terms of the rotational speed, so that the outer pins **617** all can have the same circumferential speed. In this manner, the present embodiment can reduce variation in circumferential speed among the large number of outer pins **617**.

(269) In the speed reducer **600** relating to the present embodiment, the outer pins **617** can have a larger radial dimension by having the diameter-enlarged portions **617f** without affecting the external gears (oscillating gears) **614**, which are inside the outer pins **617** in the radial direction and mesh with the outer pins **617**. This can increase the circumferential speed at the site abutting the outer peripheral surface **649b**. The present embodiment can effectively and easily control the large number of outer pins **617** such that they can rotate at the same speed.

(270) The speed reducer **600** relating to the present embodiment can produce the same effects as the speed reducers relating to the above-described embodiments.

(271) While each outer pin **617** has the diameter-enlarged portions **617f** on its respective ends in the present embodiment, each outer pin **617** may have the diameter-enlarged portion **617f** only on one of the ends.

Eleventh Embodiment

(272) The following describes a speed reducer relating to an eleventh embodiment of the disclosure with reference to the accompanying drawings. FIG. **17** is a sectional view showing the speed reducer relating to the eleventh embodiment along the axial direction. The present embodiment is different from the tenth embodiment in terms of how the carriers are divided. The other components relating to the eleventh embodiment have counterparts in the tenth embodiment, and they are assigned with the same reference signs and not described now.

(273) In the speed reducer **600** relating to the present embodiment, as shown in FIG. **17**, the first carrier (shaft flange) **618** is divided into two portions next to each other in the direction extending along the central axis **6La**. Here, the dividing plane can be oriented such that the central axis **6La** is normal to the dividing plane, and aligned with the annular surface **649a**, as indicated by the dotted line in FIG. **17**. The first carrier (shaft flange) **618** is divided into an inner shaft flange **618a** and an outer shaft flange **618b**. The inner shaft flange **618a** is arranged to enclose therein the inner space of the speed reducer **600**, and the outer shaft flange **618b** is positioned outside the speed reducer **600**.

(274) The inner shaft flange **618a** is arranged to enclose therein the inner space of the speed reducer **600**. The outer periphery of the inner shaft flange **618a** is aligned with the outer peripheral surface **649b**. The thickness of the inner shaft flange **618a** is equal to the dimension of the outer peripheral surface **649b** in the axial direction. The outer shaft flange **618b** is arranged outside the speed reducer **600**. The outer periphery of the outer shaft flange **618b** is partly aligned with the annular surface **649a**. The first carrier (shaft flange) **618** is configured such that the inner and outer shaft flanges **618a** and **618b** are stacked on each other in the direction extending along the central axis **6La**.

(275) Likewise, the second carrier (hold flange) **620** can be divided into two portions next to each other in the direction extending along the central axis **6La**. Here, the dividing plane can be oriented such that the central axis **6La** is normal to the dividing plane, and aligned with the annular surface **649a**, as indicated by the dotted line in FIG. **17**. The second carrier (hold flange) **620** is divided into an inner hold flange **620a** and an outer hold flange **620b**. The inner hold flange **620a** is arranged to enclose therein the inner space of the speed reducer **600**, and the outer hold flange **620b** is positioned outside the speed reducer **600**.

(276) The inner hold flange **620a** is arranged to enclose therein the inner space of the speed reducer **600**. The outer periphery of the inner hold flange **620a** is aligned with the outer peripheral surface **649b**. The thickness of the inner hold flange **620a** is equal to the dimension of the outer peripheral surface **649b** in the axial direction. The outer hold flange **620b** is arranged outside the speed reducer **600**. The outer periphery of the outer hold flange **620b** is partly aligned with the annular surface **649a**. The second carrier (hold flange) **620** is configured such that the inner and outer hold flanges **620a** and **620b** are stacked on each other in the direction extending along the central axis **6La**.

(277) The four flanges, specifically, the inner and outer shaft flanges **618a** and **618b**, and the inner and outer hold flanges **620a** and **620b** are all connected to the carrier pins (inner pins) **638** using the bolts **638a**.

(278) When the speed reducer **600** relating to the present embodiment is assembled, the input shaft **612** and its surrounding parts, the outer pins **617** and the external gears **614** are housed in the casing **622**. After this, the two flanges, or the inner shaft flange **618a** and the inner hold flange **620a** are incorporated such that they touch the diameter-enlarged portions **617f** of the outer pins

617.

(279) Furthermore, the carrier pins **638** are introduced into the carrier pin holes **639**, and the two flanges, or the outer shaft flange **618b** and the outer hold flange **620b** are brought into contact with or placed near the end surfaces **617b** of the outer pins **617**. Following this, the inner and outer shaft flanges **618a** and **618b**, and the inner and outer hold flanges **620a** and **620b** are coupled to the carrier pins (inner pins) **638** using the bolts **638a**.

(280) In the above manner, the speed reducer **600** can be assembled without being affected by the presence of the diameter-enlarged portions **617f** in the outer pins **617**, so that the assembling can be completed within a shorter period of time and the efficiency can be improved. Furthermore, the outer peripheral surface **649b**, which is brought into contact with the outer pins **617**, can be formed simply by controlling the shape of the outer periphery of the inner shaft and hold flanges **618a** and **620a**. This means that the flanges can be more easily made. When the carrier pins **638** are formed as an integral part of the first carrier (shaft flange) **618**, the carrier pins **638** can be integrally made with the inner shaft flange **618a**.

(281) The speed reducer **600** relating to the present embodiment can produce the same effects as the speed reducers relating to the above-described embodiments.

(282) Like the ninth embodiment, for example, the present embodiment can be practiced without the diameter-enlarged portions **617f** in the outer pins **617**.

Twelfth Embodiment

(283) The following describes a speed reducer relating to a twelfth embodiment of the disclosure with reference to the accompanying drawings. FIG. **18** is a sectional view showing the speed reducer relating to the twelfth embodiment along the axial direction. The present embodiment is different from the ninth to eleventh embodiments in terms of the main bearings. The other components relating to the twelfth embodiment have counterparts in the ninth to eleventh embodiments, and they are assigned with the same reference signs and not described now.

(284) In the speed reducer **600** relating to the present embodiment, as shown in FIG. **18**, the main bearing **624** arranged on the outer pins **617** at one end thereof is a crossed roller bearing. The other ends of the outer pins **617** serve as the main bearing **626**, as in the ninth to eleventh embodiments. In the main bearing **626**, the pin grooves **616b** extend along the entire length of the outer pins **617** extending along the central axis **6La**. The pin grooves **616b** extend from the radially outside of the external gears **614** to the radially outside of the second carrier **620**.

(285) Therefore, like the pin grooves **616b**, the casing **622** having the internal gear **616** integrated therewith extends to the radially outside of the second carrier **620**. In this way, the outer pins **617**, which constitute the internal teeth of the internal gear **616**, and the pin grooves **616b**, form the inner peripheral surface **648** of the main bearing **626**.

(286) The main bearing **624** includes an outer ring **624a** rotatable integrally with the casing **622**, an inner ring **624b** rotatable integrally with the first carrier **618**, and a plurality of rollers **624c**, which are rolling elements. A V-groove having a V-shaped cross-section is formed on the surfaces of the outer and inner rings **624a** and **624b** that face each other, so that the V grooves of the outer and inner rings **624a** and **624b** face each other. Between the outer ring **624a** and the inner ring **624b**, the rollers **624c** are arranged such that, using the V-grooves, the axes of the adjacent rollers run orthogonally to each other.

(287) The main bearing **624**, which is a crossed roller bearing, is adjacent to the outer pins **617** on the output side in the direction extending along the central axis **6La**. The output-side ends of the outer pins **617** are adjacent to the outer ring **624a**. The inner ring **624b** is, in the direction along the central axis **6La**, spaced apart from the external gear **614** that is located on the output side. The main bearing **624**, which is a crossed roller bearing, is preferably made of metals.

(288) The outer ring **624a** is mounted in a recess formed in the casing **622**. In the present embodiment, a member **650**, which is a stationary or output member, is attached to the casing **622** using bolts **650a**. The outer ring **624a** of the main bearing **624** is sandwiched between the portions

of the casing **622** to which the bolts **650a** are attached. In other words, the casing **622** and the member **650** integrated with the casing **622** cover the outer ring **624a** of the main bearing **624**. The inner ring **624b** of the main bearing **624** is arranged radially outside and attached to the first carrier **618**.

(289) In the speed reducer **600** of the present embodiment, the carrier pins **638** have a flange portion **638b** at the output-side end thereof. The output-side ends of the carrier pins **638** run through holes **618d** formed in the first carrier **618**. The output-side ends of the carrier pins **638** are fixedly secured in the first carrier **618** using the flange portions **638b**. The input-side ends of the carrier pins **638** are fitted in holes **620d** formed in the second carrier **620**. The input-side ends of the carrier pins **638** make male screw portions **638e**, which are fixedly secured in the second carrier **620** using nuts **638d**. The carrier pins **638** are surrounded by tubular inner rollers **637**. The carrier pins (inner pins) **638** are inserted into and run through the carrier pin holes (inner pin holes) **639**, which are formed in the external gears **614**, with a gap therebetween.

(290) In the present embodiment, the input shaft bearing **634** supports the output-side end of the input shaft **612**. The input shaft **612** is connected, at the input side thereof, to a motor shaft or the like, which is not shown. Here, a motor shaft or other bearings can be used to support the input shaft **612** on the input side thereof. For this reason, no bearings are shown on the input side of the input shaft **612**.

(291) In the present embodiment, the main bearing **624** on the output side is a crossed roller bearing. This can produce rolling contact and thus diminish the resistance, so that the carriers **618** and **620** can rotate relative to the casing **622** stably. In particular, during the rotation, the carriers **618** and **620** and the casing **622** can remain at a desired position more stably in the thrust direction.

(292) The speed reducer **600** relating to the present embodiment can produce the same effects as the speed reducers relating to the above-described embodiments.

Thirteenth Embodiment

(293) The following describes a speed reducer relating to a thirteenth embodiment of the disclosure with reference to the accompanying drawings. FIG. **19** is a sectional view showing the speed reducer relating to the thirteenth embodiment along the axial direction. The present embodiment is different from the twelfth embodiment in terms of the main bearings. The other components relating to the thirteenth embodiment have counterparts in the twelfth embodiment, and they are assigned with the same reference signs and not described now.

(294) In the speed reducer **600** relating to the present embodiment, as shown in FIG. **19**, the input-side ends of the outer pins **617** also serve as the main bearing **626**, as in the twelfth embodiment. In the main bearing **626**, the pin grooves **616b**, which form the inner peripheral surface **648**, do not extend along the entire length of the outer pins **617** in the direction extending along the central axis **6La**. In other words, the pin grooves **616b** are positioned radially outside the external gears **614**, but not positioned radially outside the second carrier **620**.

(295) Like the pin grooves **616b**, the casing **622** covers and is positioned radially outside the external gears **614**. The casing **622** does not occupy a region radially outside the second carrier **620**. The input-side end surface **622a** of the casing **622**, which is positioned on the inner-periphery side near the central axis **6La**, may be flush with the output-side end surface of the second carrier **620**. Accordingly, the outer pins **617** are not in contact with the pin grooves **616b** on the radially outer side with respect to the second carrier **620**.

(296) Since the casing **622** does not occupy the region radially outside the second carrier **620**, the speed reducer **600** relating to the present embodiment can achieve further reduction in weight. Furthermore, the speed reducer **600** relating to the present embodiment can produce the same effects as the speed reducers relating to the above-described embodiments.

Fourteenth Embodiment

(297) The following describes a speed reducer relating to a fourteenth embodiment of the disclosure with reference to the accompanying drawings. FIG. **20** is a sectional view showing the

speed reducer relating to the fourteenth embodiment along the axial direction. The present embodiment is different from the twelfth and thirteenth embodiments in terms of the main bearings. The other components relating to the fourteenth embodiment have counterparts in the twelfth and thirteenth embodiments, and they are assigned with the same reference signs and not described now.

(298) In the speed reducer **600** relating to the present embodiment, as shown in FIG. **20**, the input-side ends of the outer pins **617** also serve as the main bearing **626** as in the thirteenth embodiment and are not covered by the pin grooves **616b** forming the inner peripheral surface **648**. The outer pins **617** have the diameter-enlarged portions **617f** at the input-side ends thereof as in the tenth embodiment.

(299) The outer pins **617** of the main bearing **626** have the diameter-enlarged portions **617f** on the input side, where the outer pins **617** are not in contact with the pin grooves **616b**. The outer pins **617** have the diameter-enlarged portions **617f** at the portion that is on the input side in the axial direction with respect to the end surface **622a** of the casing **622**. The diameter-enlarged portions **617f** are formed in the portion of the outer pins **617** that protrudes beyond the pin grooves **616b**. The diameter-enlarged portions **617f** are sandwiched, in the direction extending along the central axis **6La**, between the end surface **622a** and the annular surface **649a**. The diameter-enlarged portions **617f** partly protrude beyond the circumferential recess **647** in the direction extending along the central axis **6La**.

(300) In the speed reducer **600** configured in the above manner, the main bearing **624**, which is positioned on the output side, is a crossed roller bearing so that it can stably keep the parts in the designed position in the thrust direction and is made of metals so that it can ensure that the casing **622** and the carriers **618** and **620** can rotate relative to each other stably. In the speed reducer **600**, the diameter-enlarged portions **617f** are positioned on the input side. The output-side end of the diameter-enlarged portions **617f** in the central axis **6La** direction is restricted from moving out of position by the end surface **622a**, and the input-side end of the diameter-enlarged portions **617f** in the central axis **6La** direction is restricted from moving out of position by the annular surface **649a**. This arrangement can contribute to restrict the casing **622** and the carriers **618** and **620** from moving out of position in the direction extending along the central axis **6La**. Accordingly, the speed reducer **600** can operate more stably.

(301) The speed reducer **600** relating to the present embodiment can produce the same effects as the speed reducers relating to the above-described embodiments.

(302) The features of the above-described embodiments of the present invention can be combined.

(303) In addition, Polyaryletherketones (PAEKs) include polyetherketones (PEKs), polyetherketone ketones (PEKKs), and polyetherketone ketone ether ketones (PEKKEKs).

(304) The present disclosure include the following inventions (1-1) to (1-6). (1-1) An eccentric oscillation speed reducer for transmitting a driving force between a first member and a second member, the eccentric oscillation speed reducer being configured to change the number of rotations at a predetermined ratio, the eccentric oscillation speed reducer comprising: an eccentric portion; an external gear having external teeth and an insertion hole into which the eccentric portion is inserted; a casing attachable to one of the first and second members; a carrier attachable to the other of the first and second members; and a main bearing having an inner sliding surface and an outer sliding surface, the inner sliding surface being rotatable integrally with the casing, the outer sliding surface being rotatable integrally with the carrier, wherein the casing has internal teeth meshing with the external teeth of the external gear, wherein the carrier is placed within the casing in a radial direction while holding the external gear, wherein the casing and the carrier are rotatable relative to each other in a concentric manner in response to oscillation of the external gear resulting from rotation of the eccentric portion, and wherein one of the inner and outer sliding surfaces is made of resin, and the other of the inner and outer sliding surfaces is made of a thermally conductive material that is more wear-resistant than the resin. (1-2) An eccentric oscillation speed

reducer including two or more eccentric bodies and external gears respectively corresponding to the eccentric bodies, the eccentric oscillation speed reducer comprising: a casing having an internal gear on an inner periphery thereof, the internal gear meshing with the external gears; a carrier rotatable relative to the casing and rotatable relative to the eccentric bodies; and a main bearing having an inner sliding surface and an outer sliding surface, the inner sliding surface being formed on an inner peripheral surface of the casing, the outer sliding surface being formed on an outer peripheral surface of the carrier, wherein one of the inner and outer sliding surfaces is made of resin, and the other of the inner and outer sliding surfaces is made of a thermally conductive material that is more wear-resistant than the resin. (1-3) A speed reducer comprising: an internal gear having a casing and a plurality of outer pins, the casing surrounding a main axis, the outer pins being rotatably arranged in pin grooves provided on an inner periphery of the casing; an external gear meshing with the internal gear; an eccentric body for oscillating the external gear; a carrier rotatable relative to the casing; and a main bearing having an inner surface and an outer surface, the inner surface being rotatable integrally with the casing, the outer surface being rotatable integrally with the carrier, the outer pins being sandwiched between the inner surface and the outer surface, wherein the inner and outer surfaces are made of one of a resin and a thermally conductive material that is more wear-resistant than the resin, and the outer pins are made of the other. (1-4) The speed reducer as set forth in (1-3), wherein the outer pins have, at an end thereof, a diameter-enlarged portion having an enlarged diameter. (1-5) A speed reducer comprising: an internal gear having a casing and a plurality of outer pins, the casing surrounding a main axis, the outer pins being rotatably arranged in pin grooves provided on an inner periphery of the casing; an external gear meshing with the internal gear; an eccentric body for oscillating the external gear; a carrier rotatable relative to the casing; a first main bearing having an inner surface and an outer surface, the inner surface being rotatable integrally with the casing, the outer surface being rotatable integrally with the carrier, the outer pins being sandwiched between the inner surface and the outer surface; and a second main bearing configured as a crossed roller bearing including a plurality of rollers disposed between V grooves having a V-shaped section facing each other, wherein the inner and outer surfaces are made of one of resin and a thermally conductive material that is more wear-resistant than the resin, and the outer pins are made of the other. (1-6) A speed reducer comprising: an internal gear having a casing and a plurality of outer pins, the casing surrounding a main axis, the outer pins being rotatably arranged in pin grooves provided on an inner periphery of the casing; an external gear meshing with the internal gear; an eccentric body for oscillating the external gear; a carrier rotatable relative to the casing; and a main bearing having an inner surface and an outer surface, the inner surface being rotatable integrally with the casing, the outer surface being rotatable integrally with the carrier, the outer pins being sandwiched between the inner surface and the outer surface, wherein the inner and outer surfaces are made of resin, and the outer pins are made of a metal, and wherein the outer pins have, at an end thereof, a diameter-enlarged portion having an enlarged diameter.

INDUSTRIAL APPLICABILITY

(305) The present disclosure can be applied to, for example, service and collaborative robots, power assist prostheses and other devices that are placed at a short physical distance from humans. The present disclosure can effectively provide safety through weight reduction.

Claims

1. A speed reducer comprising: a casing configured to surround a main axis; an internal gear integrally formed with the casing; an external gear meshing with the internal gear; an eccentric body for oscillating the external gear; a carrier rotatable relative to the casing; and a main bearing configured to rotatably support to the carrier relative to the casing, wherein the internal gear includes an internal gear body integrated into the casing, and a plurality of internal tooth pins

configured to mesh with the external gear, wherein the plurality of internal tooth pins is configured to rotatably be arranged in pin grooves formed on an inner peripheral surface of the internal gear body, wherein the main bearing functions as a plain bearing having an inner sliding surface and an outer sliding surface that are in contact with each other, wherein an inner peripheral surface of a metal ring being rotatable integrally with the casing functions as the inner sliding surface, wherein an outer peripheral surface of the carrier functions as the outer sliding surface, wherein the casing and the carrier are made of resin, wherein, in a direction extending along the main axis, the metal ring and the plurality of internal tooth pins is positioned to overlap each other, wherein the internal gear body is made of resin, and wherein the plurality of internal tooth pins, the metal ring, and the carrier are made of a material that has a higher thermal conductivity than the resin of the internal gear body.

2. The speed reducer of claim 1, wherein the inner and outer sliding surfaces have a groove formed therein that is not in communication with an internal space having the external gear housed therein.
